

Semantic Matching in Medieval Latin: A Benchmark, Layerwise Analysis, and Geometric Repair

Anonymous ACL submission

Abstract

Medieval canon law before the twelfth century survives in many compilations, whose manuscripts differ in spelling and in the selection and arrangement of their elements. We introduce a Medieval Latin benchmark of 840 expert-labelled canon-law fragments with 1,705 manuscript witnesses, and we define two tasks on it: pairwise duplicate detection and open-set directory routing. Our question is where inside a pre-trained encoder the retrieval signal lives, why it disappears, and what recovers it. A layerwise analysis of six pre-trained models from three architectures separates two effects that are easy to conflate. Anisotropy is universal across the six models: every model has a layer where the mean cosine between randomly selected passages reaches 0.23 to 0.95, and no baseline layer in any model separates equivalent from non-equivalent passages by more than 0.29 in mean cosine. A mid-depth collapse of retrieval is not universal: the Latin and multilingual T5 encoders fall to AUROC 0.496, 0.538, and 0.654 in their middle layers, while LaBSE, Qwen3-0.6B, and KaLM-mini improve close to monotonically with depth. All-but-the-Top post-processing, fit only on training embeddings, repairs all six. Best-layer AUROC moves into a 0.971 to 0.987 band, the cosine gap rises to between 0.525 and 0.611, and the spread in directory accuracy at rank 1 across models falls from 39.3 points to 3.3. Under the repair, integrated gradients and retrieval-adapted MaRC give a narrow answer: ABTT improves leave-one-out rank faithfulness in five of six LaTa, PhilTa, and mT5-base model-method cells, the sixth a tie, while ERASER-style rationale metrics show mixed effects.

1 Introduction

Medieval canon law preserves early records of legislation and judicial opinions in Western Europe. Before the twelfth century it survives not as

one stable text but in many compilations, whose manuscripts differ in spelling, wording, fragment boundaries, and transmission history. Deciding whether two passages express the same legal content despite those surface changes drives editorial work: identifying duplicates, analyzing witnesses, and tracing how a text was transmitted.

Identifying the source of such a passage is expert work. Scribes often left the source out or named the wrong one, so each passage has to be traced to the council, pope, ruler or church father it comes from, and that tracing relies on specialised reference works that few scholars command. The several hundred manuscripts behind the Carolingian Canon Law project (Firey, 2009) still hold thousands of passages with no identified source (Appendix A). A system that ranks candidate sources leaves the identification to the expert and is meant to shorten the search that precedes it. Each passage identified this way becomes dated, located evidence for historians of law, theology and language.

We study this problem as semantic matching on a benchmark of 840 expert-labelled fragments of Medieval Latin canon law, each a canon or excerpt with one or more manuscript witnesses, 1,705 in all. Task A asks whether two passages express the same meaning. Task B turns the same embeddings into an open-set workflow: route a new passage to an existing directory, or mark it as new. Our question is where inside a pre-trained encoder the retrieval signal lives, why it disappears, and what recovers it.

A final-layer vector cannot answer that, so we work layer by layer, and the layerwise view of six models separates two effects that are often reported together. Anisotropy is universal on this corpus: every model has a layer where pooled passages crowd into a narrow cone, with mean cosine between randomly selected passages reaching 0.23 to 0.95, and no baseline layer of any model separates equivalent from non-equivalent passages by more than 0.29

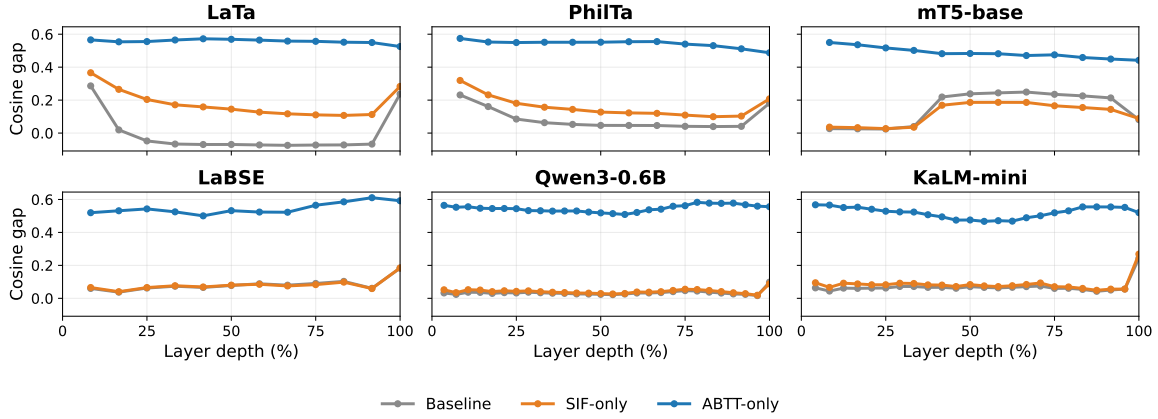


Figure 1: Cosine gap against layer depth for all six models: mean cosine of same-directory pairs minus mean cosine of different-directory pairs, the margin a threshold works with, so larger is better. Gray is baseline mean pooling, blue is ABTT-only with D tuned on train, orange is SIF-only. Baseline stays under 0.29 at every layer of every model, and under 0.11 for Qwen3-0.6B. Mid-depth in the top row, the LaTa and PhilTa baseline gap collapses toward zero with their retrieval; the mT5-base gap rises as its ranking fails, which only AUROC (Figure 2) shows. ABTT lifts all six near 0.5 at every depth. SIF-only helps LaTa and PhilTa and leaves the other four near baseline.

in mean cosine (Figure 1). A catastrophic mid-depth collapse of retrieval is not universal. LaTa and PhilTa fall to chance in their middle layers and mT5-base loses more than half of its above-chance separation, while LaBSE, Qwen3-0.6B, and KaLM-mini improve close to monotonically with depth.

Anisotropy motivates our use of All-but-the-Top (ABTT) post-processing (Mu and Viswanath, 2018). We apply it after pooling, with no update to model parameters and with principal components fit on training embeddings only. The repair, unlike the collapse, is universal: ABTT lifts every model into a narrow high-accuracy band on both tasks, and at each T5 encoder’s worst baseline layer it recovers pairwise AUROC to within 1.0 point of that model’s best ABTT layer. SIF weighting (Arora et al., 2017), which reweights tokens by frequency before pooling, corrects part of the same direction but helps only the collapsed models, and SIF-only never matches ABTT-only.

Geometry diagnoses collapse and recovery but does not select the retrieval layer: a train-only rule (Section 5) locks the layers before any attribution metric is computed.

Finally, we compare local explanations of the score before and after the repair. Integrated gradients (Sundararajan et al., 2017) gives a gradient view of token importance and retrieval-adapted MaRC (Brinner and Zarri  , 2023) a learned-mask view; both explain a fixed query-candidate score and stay local diagnostics.

This paper makes four contributions: (1) we

introduce an expert-labelled benchmark for semantic matching in Medieval Latin canon law, built on the source keys assigned by the Carolingian Canon Law project, together with a routing task that states the scholar’s decision directly: file an unidentified witness under a known fragment, or flag it as new; (2) we separate, across six models, universal anisotropy from the mid-depth retrieval collapse that only the three T5 encoders show, measuring both with top-PC dominance, effective rank, and cosine concentration; (3) we show that ABTT repairs the anisotropy on this corpus without updating parameters, compressing a 39.3-point spread in directory accuracy at rank 1 to 3.3 points, with the largest gains on the T5 encoders; and (4) we adapt token attribution to bi-encoder retrieval, comparing a MaRC-style learned mask with integrated gradients at predeclared retrieval-selected layers; on present evidence ABTT improves leave-one-out rank faithfulness in five of six model-view cells, the sixth a tie, while ERASER-style rationale metrics stay mixed.

2 Related Work

Anisotropy, layer geometry, and repair. Pre-trained spaces occupy narrow cones (Gao et al., 2019; Ethayarajh, 2019) where a few coordinates dominate cosine (Kovaleva et al., 2021; Timkey and van Schijndel, 2021), and corrections remove or rescale those directions (Mu and Viswanath, 2018; Su et al., 2021) or downweight frequent tokens (Arora et al., 2017). Concurrent work on nineteen encoders’ output embeddings finds that re-

moving dominant directions restores cosine where the space is anisotropic (Parupudi, 2026); we show the effect at every depth and separate it from the T5 retrieval collapse. Information spreads across depth (Tenney et al., 2019; Liu et al., 2019), intermediate layers can beat the last (Skean et al., 2025), and multilingual work is rarer (Rajae and Pilehvar, 2022). Layerwise profiles put the mid-depth anisotropy peak in decoder-only models (Razhigaev et al., 2024); T5 enters that work with encoder and decoder states concatenated (Godey et al., 2024), and its weak mean-pooled similarity is a final-layer result (Ni et al., 2022). Anisotropy is absent in some model families (Machina and Mercer, 2024) and is not always harmful (Ait-Saada and Nadif, 2023); on this corpus it appears in all six models we test. To our knowledge, no prior profile reports layerwise retrieval quality for T5 encoder states in isolation, or applies a top-component correction at every depth. We do both, and find that the geometric peak and the retrieval dip dissociate.

Latin and classical-language NLP. Latin has dedicated encoders and encoder-decoders (Bamman and Burns, 2020; Riemenschneider and Frank, 2023a), and appears in multilingual and decoder-only models (Xue et al., 2021; Feng et al., 2022; Zhang et al., 2025; Hu et al., 2025). Shared tasks target tagging (Sprugnoli et al., 2020), retrieval work targets allusion (Riemenschneider and Frank, 2023b), and charter work semantic change (Liu et al., 2025). General retrieval benchmarks (Thakur et al., 2021; Muennighoff et al., 2023) contain no Latin retrieval task, and concurrent Latin reuse benchmarks target classical and patristic literature (Schelb et al., 2026; de la Selle, 2026). We contribute a benchmark of medieval canon-law manuscript witnesses, with open-set routing and a layerwise analysis, labelled from an existing scholarly edition.

Text reuse in digital humanities. Text-reuse systems match n-grams (Coffee et al., 2012), align reprinted newspapers (Smith et al., 2013), and add distributional semantics for allusive reuse (Manjavacas et al., 2019; Burns et al., 2021), ranking candidates in a closed collection. Our routing task is open-set: a witness may match no labelled fragment in the collection.

Faithfulness of token attributions. Integrated gradients (Sundararajan et al., 2017) and learned masks (Brinner and Zarrieß, 2023) score tokens,

and their faithfulness is debated (Adebayo et al., 2018; Jacovi and Goldberg, 2020). Erasure protocols (DeYoung et al., 2020; Atanasova et al., 2020) target classification and retrieval explanation targets ranking (Anand et al., 2022); we apply both to a bi-encoder cosine score.

3 Task and Dataset

3.1 Task Definition

The benchmark defines two tasks over the same witnesses and the same pooled embeddings.

Task A: pairwise duplicate detection. Let p_1 and p_2 be two passages and let $\equiv$ denote semantic equivalence. The label is $y = 1$ if $p_1 \equiv p_2$ and $y = 0$ otherwise. The model scores the pair by $\cos(e_{p_1}, e_{p_2})$ over pooled embeddings. Task A uses no directory structure and no threshold at scoring time, and evaluation uses the test $n \times n$ cosine matrix.

Task B: directory routing. Given a query passage q and the existing labelled directories, the system routes q to a directory or marks q as new. In autonomous routing, a threshold τ fit on training same-directory and different-directory pairs decides whether the query enters its nearest neighbour’s directory or the new class; in top- k ranking, each directory receives its best document match score, a pseudo-option `__NEW__` receives score τ , and the simulated expert succeeds when the labelled directory appears in the shortlist.

3.2 Dataset

The material comes from the Carolingian Canon Law project (CCL; Firey, 2009), an open-access digital edition of early medieval canon-law manuscripts, most of them ninth-century and never edited to modern standards. The project transcribes the manuscripts, almost all by hand, and gives each transcribed canon or excerpt two identifiers: a location identifier (manuscript siglum, folium, position on the folium) and, where the source is identified, a source key naming the conciliar canon, papal decretal, royal capitulary or patristic excerpt that the unit copies. Keys are assigned by the project’s director or by graduate students under the director’s supervision, who have enough Latin to check the manuscript text against the modern authoritative version of the source. The project verifies and occasionally corrects keys through its parallel display

Category	Dirs	Train	Test
Singletons (1 file)	545	272	273
Doubletons (2 files)	108	108	108
Multi-file (≥ 3)	187	467	477
Total	840	847	858

Table 1: Train and test split statistics. Doubleton directories are assigned as whole folders. The test set contains 535 existing files, with a same-directory partner in test, and 323 new files.

of all units that share a key, where a misidentified unit stands out against the others.

The labels are therefore expert-assigned and checked by collation, and they are the project’s work, as are the transcriptions. The benchmark is ours: we take the units that carry a key, log the label corrections the project supplied, freeze the file set under a checksum manifest, and define the leak-free split and the two tasks. A directory is one labelled fragment, named by its source key, and a file is one manuscript witness of that fragment (one manuscript’s copy of it), held as plain text derived from the project’s TEI XML exports. Appendix A gives the transcription workflow, the identifier design, the label taxonomy and the derivation rule.

The benchmark holds 1,705 witnesses in 840 directories, with directory sizes from one to ten. Witnesses of one fragment are free to vary from each other, so the benchmark asks whether an unidentified witness belongs to a known fragment: files in the same directory count as positive matches, files from different directories count as negatives, and singleton directories contribute no positive pairs.

3.3 Train/Test Split

We split the corpus into 847 training files and 858 test files with a leak-free protocol (Table 1). The 545 singleton directories are distributed at the file level. Doubleton directories are assigned as whole folders, 54 to train and 54 to test, so both files from a train doubleton contribute a positive training pair. For directories with at least three files, the train allocation varies inside each size class while keeping each class near a balanced file split, which avoids a rule where every size-three directory contributes no positive training pairs. The final split contains 565 positive training pairs and 596 positive test pairs.

4 Methodology

Our pipeline has three stages: pool token states from one layer into a passage vector, correct the geometry, then score pairs by cosine.

4.1 Models

We evaluate six pre-trained models from three architectures. LaTa is a monolingual Latin T5 and PhilTa a trilingual (Ancient Greek, Latin, English) T5, both released by [Riemenschneider and Frank \(2023a\)](#), each with 12 encoder layers, and mT5-base is a multilingual T5 encoder-decoder, also with 12 encoder layers ([Xue et al., 2021](#)). LaBSE is a language-agnostic BERT sentence encoder covering 109 languages, with 12 encoder layers ([Feng et al., 2022](#)). Qwen3-0.6B ([Zhang et al., 2025](#)) and KaLM-mini ([Hu et al., 2025](#)) are multilingual decoder-only embedding models with 28 and 24 layers.

4.2 Embedding Extraction

We extract hidden-state representations from every transformer block, starting at layer 1 and skipping layer 0, the static embedding table. Token representations at each layer are pooled into one passage vector by plain mean pooling or SIF weighting ([Arora et al., 2017](#)). Appendix E compares last-token pooling, a common choice for decoder-only models, with mean pooling under ABTT.

The paper-facing pooling baseline removes only tokenizer artifacts with no lexical content, such as whitespace, padding, and tokenizer-prefix-only tokens; punctuation and numerals remain in the pooled representations.

4.3 Post-Processing

ABTT. Our main post-processing method is All-but-the-Top (ABTT) ([Mu and Viswanath, 2018](#)). For each model and layer, we center the pooled training embeddings, compute the top D principal components by SVD, and remove those components from both train and test embeddings.

SIF. SIF weighting ([Arora et al., 2017](#)) divides each token weight by its estimated corpus frequency, down-weighting high-frequency tokens. SIF and ABTT act on the same problem at different stages: SIF reweights tokens before pooling, ABTT removes dominant directions after pooling.

PCA whitening (excluded). PCA whitening ([Su et al., 2021](#)) rescales principal directions rather than only removing the top components; on this corpus it degenerates the learned routing threshold, so we report its sweep only in Appendix G.

4.4 Local Attribution Methods

We evaluate local explanations for a fixed query-candidate retrieval score. The design goal is a review aid: attributions are shown as highlights connecting matching tokens across the two witnesses, so a scholar judging a proposed match can see which words the score rests on. Integrated gradients (Sundararajan et al., 2017) gives a gradient-based attribution of pairwise cosine similarity to input tokens. We compute per-sequence token attributions for each side of the pair (pair-matrix view: Figure 5).

Our learned-mask attribution is adapted from MaRC (Brinner and Zarrieß, 2023). MaRC was proposed for classification, where a soft input mask preserves a target class score while sparsity and smoothness regularizers encourage concise rationales. We keep the per-instance mask-optimization idea and replace the classification target with pairwise retrieval cosine. The mask is optimized on one side of a query-candidate pair while the partner representation stays fixed, and swapping the two sides gives candidate-side attribution.

Let $\mathbf{E}(\mathbf{x}_q) \in \mathbb{R}^{T_q \times d_{\text{emb}}}$ denote the input-embedding lookup, $E_{>0}$ the encoder with the lookup removed, and $\mathbf{z}_c^*$ the candidate representation held fixed during optimization. We parameterize $\mathbf{m} = \sigma(\boldsymbol{\zeta})$ with free logits $\boldsymbol{\zeta} \in \mathbb{R}^{T_q}$ and write $\mathcal{I}$ for content-token positions. For the ABTT variant, define $s_{\text{ABTT}}(\mathbf{a}, \mathbf{b}) = \cos(\text{ABTT}(\mathbf{a}), \mathbf{b})$. The retrieval-adapted MaRC loss is

$$\begin{aligned} \mathcal{L}(\mathbf{m}) = & (s_{\text{ABTT}}(\tilde{\mathbf{z}}_q(\mathbf{m}), \mathbf{z}_c^*) - s_{\text{ABTT}}(\mathbf{z}_q, \mathbf{z}_c^*))^2 \\ & + \lambda \sum_{i \in \mathcal{I}} \frac{m_i}{|\mathcal{I}|} + \gamma \sum_{i, i+1 \in \mathcal{I}} \frac{|m_{i+1} - m_i|}{|\mathcal{I}| - 1}, \end{aligned} \quad (1)$$

where $\tilde{\mathbf{z}}_q(\mathbf{m}) = \text{pool}(E_{>0}(\text{diag}(\mathbf{m}) \mathbf{E}(\mathbf{x}_q)), \mathbf{a}_q)$. The baseline variant uses raw cosine in place of the ABTT-corrected cosine in Equation 1. Logits outside $\mathcal{I}$ are frozen at $-\infty$.

5 Experimental Setup

All fitting uses training data only: SIF token probabilities, ABTT principal components, routing thresholds, D , and the operational layer rule. We train no model parameters, so the protocol uses no development split, and every choice is evaluated once on test. The one exception, the supervised fine-tuning ceilings of Tables 2 and 3, carve their development slice out of the training files by directory, never out of test (Appendix H). We report

four settings throughout: baseline mean pooling, SIF-only, ABTT-only, and SIF+ABTT.

Task A metrics. AUROC treats same-directory test pairs as positives and different-directory pairs as negatives. Cosine gap is the same-directory minus the different-directory mean cosine.

Task B metrics. For autonomous routing we report assignment, existing, and new accuracy. Assignment accuracy scores only the existing-versus-new decision: a file with a partner in test is correct when its maximum cosine to another test file is at least τ , and a file without one is correct when that maximum falls below τ . It never checks which directory the file would enter, so a degenerate threshold that routes everything as existing already attains the 62.4 percent class prior; read it next to directory accuracy at rank 1, which does require the correct directory. For top- k ranking, cumulative top- K accuracy is the fraction of test queries whose labelled directory is among the top K options.

Attribution metrics. We report two threshold-free metrics: ρ_{LOO} , the Spearman correlation between attribution magnitude and the leave-one-out drop in the same retrieval score, and the DeLAUC gap, the random-order minus attribution-order deletion-curve area. ERASER-style sufficiency, comprehensiveness, and MinFrac (DeYoung et al., 2020), defined for retrieval cosine in Appendix I, are dropped from the main table because the sufficiency side fails our shuffled-attribution control and their verdicts depend on the threshold and on the erasure operator. Every curve metric ranks tokens by $|a|$, ties broken by position; Appendix I reports them and states both conventions and their consequences.

Hyperparameters and layer selection. We search $D \in \{1, 2, 3, 5, 7, 10\}$ and choose D by training-set Task B directory accuracy at rank 1, and MaRC runs up to 200 Adam steps per pair (learning rate 0.1, stopping after step 50 once the masked cosine is within 0.01 of the full one) from $\zeta_0 = \log 9$ with $\lambda = 0.01$ and $\gamma = 0.001$; integrated gradients uses 40 integration steps. Label-free geometry diagnostics (top-PC dominance, effective rank, cosine concentration) identify the most anisotropic layers and test whether ABTT repairs them. For retrieval and attribution we apply a predeclared train-only rule: choose the earliest layer within 0.5 points of the best training-set directory accuracy at rank 1 under ABTT. It selects LaTa

layer 7, PhilTa layer 1, and mT5-base layer 1 for attribution, with $D = 10$ ABTT components. The most anisotropic layers are LaTa layer 8, PhilTa layer 6, and mT5-base layer 5, kept as mechanism checks.

6 Results

6.1 Task A: Pairwise Duplicate Detection

In Table 2, baseline quality is uneven, with AUROC from 0.838 for mT5-base to 0.972 for KaLM-mini, and ABTT removes most of that unevenness: every model lands between 0.971 and 0.987, and the cosine gap rises from a baseline range of 0.024 to 0.237 to a range of 0.525 to 0.611. The gain is largest where the baseline is weakest: mT5-base gains 0.137 AUROC, LaTa 0.034, and KaLM-mini 0.009.

The cosine gap column explains why a strong AUROC is not enough: KaLM-mini has the best baseline AUROC, 0.972, but a gap of only 0.056, and Qwen3-0.6B ranks nearly as well, 0.966, with the smallest gap of all, 0.024, so both leave almost no margin for a threshold. ABTT raises every model’s gap by at least 0.28, the margin a single learned threshold needs in Task B.

SIF as a frequency-based complement. Tables 2 and 3 show how far SIF, the frequency side of the same dominant direction, gets. On Task B assignment accuracy, SIF-only adds 9.4 points for LaTa, 8.2 for PhilTa, and 7.3 for mT5-base, the three models whose retrieval collapses; it adds 0.1 point for LaBSE and subtracts 1.9 for KaLM-mini and 6.3 for Qwen3-0.6B. SIF-only never matches ABTT-only on either task for any model, and adding SIF before ABTT moves best-layer assignment accuracy by at most 1.8 points either way. SIF is a partial account of the anisotropy, not a competing repair; the full sweeps are in Appendix G.

6.2 Layerwise Geometry Diagnosis

The layerwise profile separates the two effects: the collapse belongs to the T5 encoders. Baseline AUROC bottoms out at 0.496 for LaTa at layer 6, which is chance, at 0.538 for PhilTa at layer 10, and at 0.654 for mT5-base at layer 5, while ABTT at those layers gives 0.964, 0.982, and 0.978. For the other three, baseline AUROC rises with depth to a final-layer AUROC of 0.956 for LaBSE, 0.958 for Qwen3-0.6B, and 0.956 for KaLM-mini, with layer minima of 0.806, 0.858, and 0.861. Figure 1

plots the cosine gap across depth and Figure 2 (Appendix B) plots AUROC over the same panels.

Anisotropy, in contrast, is present in all six. At each model’s most anisotropic layer, the layer with the largest top-PC variance share, the mean pairwise cosine on the test split is 0.230 for LaTa, 0.584 for PhilTa, 0.314 for mT5-base, 0.878 for LaBSE, 0.953 for Qwen3-0.6B, and 0.887 for KaLM-mini. ABTT with $D = 10$ drives all six to within 0.002 of zero and raises effective rank from as low as 1.00 to between 147.50 and 176.36 (Appendix B). The T5 encoders concentrate variance in one direction, with top-PC shares of 0.956, 0.862, and 1.000, so the cone is low-rank and retrieval dies with it. The non-T5 models hold top-PC shares of 0.506, 0.167, and 0.396 but still carry a large common component, so passages sit close together without losing their relative order. ABTT removes that common component in both cases, which is why the repair generalizes even though the symptom does not.

At PhilTa’s collapsed retrieval layer, equivalent and non-equivalent cosines overlap heavily before ABTT and separate after it (Figure 3, Appendix C).

In t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018) projections at the train-selected layers (Appendix K), ABTT-only with $D = 10$ forms tighter same-directory groups than baseline mean pooling: the t-SNE silhouette of the 1,160 winnable witnesses rises by 0.07 to 1.16 on the three T5 encoders.

6.3 Task B: Directory Routing

In Table 3, directory accuracy at rank 1 spans 39.3 points under the baseline, from 46.6 for mT5-base to 85.9 for KaLM-mini, so the model choice dominates. ABTT compresses that to 3.3 points, from 86.1 to 89.4. Assignment accuracy tracks it, moving from a 40.3-point spread to the range 88.5 to 91.7, with every ABTT cell above every baseline cell.

Tables 14 and 17 in Appendix G repeat the top-1 comparison under five query and reference reseeds at each method’s own train-selected layer (the Task B Base and SIF+ABTT layers of Table 6), so the compression is not an artifact of one draw. The baseline spread of 36.6 points, from 50.7 for mT5-base to 87.3 for KaLM-mini, falls to 1.7 points under SIF+ABTT, from 88.9 for LaTa to 90.6 for LaBSE, with standard deviations of at most 1.0. Table 14 gives the shortlist view (per-layer: Appendix D): every model exceeds 95% by Top-2 and gains little after Top-3.

Model	Task A AUROC				Task A cosine gap			
	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	0.938	0.948	0.971	0.966	0.237	0.366	0.525	0.580
PhilTa	0.939	0.955	0.983	0.982	0.231	0.319	0.540	0.573
mT5-base	0.838	0.845	0.975	0.974	0.081	0.036	0.536	0.519
LaBSE	0.956	0.956	0.987	0.976	0.183	0.184	0.611	0.580
Qwen3-0.6B	0.966	0.942	0.973	0.972	0.024	0.053	0.553	0.540
KaLM-mini	0.972	0.964	0.981	0.981	0.056	0.056	0.568	0.535
LaTa (fine-tuned)	0.984	–	0.970	–	0.387	–	0.548	–
Qwen3-0.6B (fine-tuned)	0.996	–	0.994	–	0.684	–	0.716	–
KaLM-mini (fine-tuned)	0.997	–	0.994	–	0.636	–	0.651	–

Table 2: Task A pairwise duplicate detection for all six models under four post-processing settings. Each cell is a test-set score at the layer chosen by training-set AUROC, listed in Table 6; ABTT uses D tuned on train. Baseline AUROC spans 0.838 to 0.972, and ABTT lifts every model into a 0.971 to 0.987 band. Cosine gap is defined in Figure 1. Per-layer grids: Appendix Tables 7, 11, and 15. Below the rule, reference systems on the same split with the same evaluation code: LaTa, Qwen3-0.6B, and KaLM-mini fine-tuned contrastively on 499 of the 565 positive train pairs (a 28-directory dev curve takes the rest), a ceiling at this training budget (setup: Appendix H). ABTT moves the fine-tuned encoders’ AUROC 0.984 to 0.970 for LaTa (gap 0.387 to 0.548), 0.996 to 0.994 for Qwen3-0.6B (gap 0.684 to 0.716), and 0.997 to 0.994 for KaLM-mini (gap 0.636 to 0.651).

Model	Assignment accuracy				Directory accuracy @1			
	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	73.8	83.2	88.5	90.2	72.1	81.8	86.1	88.2
PhilTa	70.9	79.0	91.1	91.4	69.3	77.7	88.3	88.9
mT5-base	47.2	54.5	90.3	90.3	46.6	54.1	88.5	87.5
LaBSE	83.3	83.4	90.8	89.6	81.5	81.9	88.5	86.9
Qwen3-0.6B	82.3	76.0	91.5	90.8	80.3	74.2	89.4	88.6
KaLM-mini	87.5	85.7	91.7	90.8	85.9	84.0	89.4	88.0
LaTa (fine-tuned)	83.4	–	87.8	–	81.6	–	85.2	–
Qwen3-0.6B (fine-tuned)	91.4	–	92.1	–	90.8	–	91.4	–
KaLM-mini (fine-tuned)	92.4	–	93.4	–	91.7	–	92.5	–

Table 3: Task B autonomous routing for all six models under four post-processing settings, in percent. Each cell is a test-set score at the layer chosen by training-set directory accuracy at rank 1, listed in Table 6. Assignment accuracy scores the existing-versus-new decision alone, comparing each file’s maximum cosine against the train-fit threshold τ ; a degenerate threshold that routes everything as existing already attains the 62.4 percent class prior. Directory accuracy at rank 1 requires the correct directory, so read the two columns together. Per-layer grids: Appendix Tables 8, 12, and 16. Below the rule, reference systems on the same split with the same evaluation code: LaTa, Qwen3-0.6B, and KaLM-mini fine-tuned contrastively on 499 of the 565 positive train pairs (a 28-directory dev curve takes the rest), a ceiling at this training budget (setup: Appendix H). The fine-tuned encoders with ABTT sit below every zero-shot ABTT cell for LaTa (87.8 and 85.2), above every zero-shot ABTT cell for Qwen3-0.6B (92.1 and 91.4), and above every zero-shot ABTT cell for KaLM-mini (93.4 and 92.5), against cells spanning 88.5 to 91.7 and 86.1 to 89.4.

The parameter-free repair matches contrastive fine-tuning on LaTa routing, but supervision clears it on Qwen3-0.6B and KaLM-mini. Fine-tuning LaTa on the positive train pairs (499 after a directory-level dev curve; setup in Appendix H) gives an uncorrected Task A AUROC of 0.984, above every zero-shot baseline row (0.838 to 0.972), and 83.4 assignment accuracy; with ABTT on top it reaches 87.8 assignment accuracy and 85.2 directory accuracy at rank 1, below every zero-shot ABTT cell (88.5 to 91.7 and 86.1 to 89.4). ABTT also lowers that row’s AUROC from 0.984 to 0.970 while widening its cosine gap from 0.387 to 0.548,

consistent with fine-tuning having already removed part of the common component: the repair is not free on every row, and it matches what the available supervision buys on routing at this training budget. The same recipe on the two encoders we also fine-tuned does not land in the same place. Qwen3-0.6B reaches 92.1 assignment accuracy and 91.4 directory accuracy at rank 1 with ABTT, and KaLM-mini 93.4 and 92.5, both above every zero-shot ABTT cell. Over five reseeds of the Task B query and reference assignment the fine-tuned encoder with ABTT beats its own zero-shot ABTT row by 1.8 points on Qwen3-0.6B (0.923 ± 0.002

Model	Method	ρ_{LOO}		DelAUC gap	
		base	ABTT	base	ABTT
LaTa	IG	0.003	0.370	0.859	0.181
	MaRC	0.083	0.538	0.524	0.327
PhilTa	IG	0.329	0.614	0.113	0.403
	MaRC	0.469	0.445	0.201	0.313
mT5-base	IG	0.143	0.686	0.061	0.561
	MaRC	0.257	0.504	0.042	0.463

Table 4: Attribution faithfulness at the predeclared operational layers, 200 positive pairs per model, for integrated gradients (IG) and retrieval-adapted MaRC. ρ_{LOO} correlates attribution magnitude with the leave-one-out change in the cosine. DelAUC gap is the random-order minus attribution-order deletion-curve area, positive when attribution beats chance; the reference runs from 0.692 to 0.961 here, so zero is chance. Higher is better in both; boldface marks the better variant. ABTT wins 5/6 and 4/6. Ties, within 2 standard errors of zero: ρ for PhilTa MaRC. Ratio metrics are undefined below a full-query cosine of 0.05, so the DelAUC columns average 195 to 199 ABTT pairs against 199 to 200 baseline pairs. Cross-variant comparisons are descriptive. Secondary metrics: Table 19.

against 0.905 ± 0.004) and by 2.1 points on KaLM-mini (0.930 ± 0.005 against 0.909 ± 0.006), while on LaTa the two are 0.877 and 0.877. So the match is a statement about LaTa, not about post-processing in general, and it does not survive on the model that owns the best zero-shot cell in Table 3. All three ceilings are also flattered, because 206 of the 535 test queries whose directory already exists (38.5 percent) sit in a directory that supplied a training pair, and witnesses inside a directory are near-duplicates, so read each margin as an upper bound on itself.

6.4 Attribution at Retrieval-Selected Layers

Table 4 compares baseline and ABTT attributions on 200 positive query-candidate pairs per model for LaTa, PhilTa, and mT5-base. ABTT repairs the scores being explained at these layers: LaTa layer 7 moves from AUROC 0.498 to 0.962 and directory accuracy 0.302 to 0.868, PhilTa layer 1 from 0.939 to 0.977 and 0.693 to 0.883, and mT5-base layer 1 from 0.822 to 0.979 and 0.451 to 0.885.

ABTT improves rank faithfulness in five of six model-view cells (ρ_{LOO} 5/6, Kendall τ_b agreeing in the same five) by 10.3 to 22.5 standard errors, the sixth a tie at 1.5; chance-corrected deletion faithfulness improves in four of six, and both LaTa cells favour the baseline, at 2.1 and 8.5 standard errors. The sufficiency-side metrics fail our shuffled-attribution control in a baseline cell and stay in the

appendix, so we claim better-ranked rationales, not a gain on every axis.

Masking acts on cached layer- L hidden states, so every faithfulness statement is about the pooled read-out at that layer, not the encoder’s input-to-output map. Appendix I carries τ_b , chance-corrected insertion faithfulness, the ERASER-style rationale metrics and their threshold sweeps (Table 19, Figure 4); Appendix J shows one PhilTa retrieval case, with its integrated-gradient pair matrix before and after ABTT.

7 Discussion

Six models across three architectures narrow what we can claim. Anisotropy is universal on this corpus and so is the repair, but the mid-depth retrieval collapse is not. Layerwise profiles place the mid-depth anisotropy peak in decoder-only models and do not separate encoder-decoder encoders (Razhigaev et al., 2024; Godey et al., 2024), so, to our knowledge, this is the first report of the collapse for T5 encoders, and we do not read it as a property of the architecture. A single-architecture study could report that collapse as the general finding.

With a T5 encoder the middle layers are unusable without correction, so layer choice matters. With a non-T5 model the final layers already work, and ABTT still adds 3.5 to 9.1 points of directory accuracy at rank 1.

Geometry diagnoses the failure mode but does not select layers: it locates where representations collapse and recover, a train-only task signal picks retrieval layers, and the selected layers are not the most anisotropic ones. The attribution results at those locked layers need a narrow reading: ABTT changes the cosine being explained, so only the leave-one-out rank gain is broadly stable, and even it ties in one cell.

The repair also changes what a scholar sees. Under SIF+ABTT every model puts the correct directory in a two-item shortlist for more than 95% of test queries, with token attributions ranking the words the score rests on. The existing-versus-new call is still wrong for roughly one file in ten, so the open-set decision stays with the scholar. All six encoders share the geometry. Only the T5 encoders let it cost them retrieval, and one SVD per layer, fit on 847 training files, removes that cost.

Limitations

The study covers one domain, medieval canon law, and one language, Latin. The 1,705-witness corpus is large for this field, but small by current NLP standards. We test six models; others may behave differently. The mid-depth retrieval collapse we document holds for the three T5 encoders in our set, and we cannot separate architecture, pre-training objective, and checkpoint as its cause. ABTT also needs enough text to estimate principal components, so smaller archives need separate study.

Task B simulates a human-in-the-loop workflow by counting a query as successful when its labelled directory appears in the shortlist. That measures ranking quality, not the full judgment load a working scholar carries. Our attribution analysis is also local to fixed query-candidate scores. It does not explain model behavior globally, and it does not replace expert reading of retrieved passages. The consistent attribution effect is improved leave-one-out rank faithfulness, in five of six cells with the sixth a tie; other rationale metrics remain threshold-dependent and should not be reported as a broad attribution-quality gain.

The labels are expert-assigned, but by one project. Its director or supervised graduate students assigned the keys, and verification is by collation in the project’s parallel display, not by independent double annotation, so no inter-annotator agreement figure exists for this benchmark. Collation compares the witnesses of one key, so it has nothing to compare when a key has a single witness, and 545 of the 840 directories hold one witness in this benchmark.

Ethical Considerations

The benchmark holds early medieval legal texts, transcribed from manuscripts and early printings, and contains no personal data. The Carolingian Canon Law project publishes its material under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 licence (CCL, 2026a). The project has confirmed in writing that the licence covers the transcriptions, annotations and markup in the exports we use and that, in the view of the project’s Advisory Board, attribution to the project suffices; the project’s interface credits the transcriber and proofreader of each unit. The project’s corpus keeps growing and being corrected, so benchmark version 1 is a dated snapshot, fixed by a sha256 manifest of its 1,705 labelled files, and every num-

ber in this paper comes from it. We will release it under the same licence, in coordination with the project’s periodic versioned exports, and later versions will follow those exports. The intended use is research on retrieval and on assisted source identification: the system ranks candidates for an expert to judge and does not identify sources on its own.

References

- Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. 2018. [Sanity checks for saliency maps](#). In *Advances in Neural Information Processing Systems 31 (NeurIPS)*.
- Mira Ait-Saada and Mohamed Nadif. 2023. [Is anisotropy truly harmful? A case study on text clustering](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 1194–1203. Association for Computational Linguistics.
- Avishek Anand, Lijun Lyu, Maximilian Idahl, Yumeng Wang, Jonas Wallat, and Zijian Zhang. 2022. [Explainable information retrieval: A survey](#). *arXiv preprint arXiv:2211.02405*.
- Sanjeev Arora, Yingyu Liang, and Tengyu Ma. 2017. [A simple but tough-to-beat baseline for sentence embeddings](#). In *Proceedings of the 5th International Conference on Learning Representations (ICLR)*.
- Pepa Atanasova, Jakob Grue Simonsen, Christina Lioma, and Isabelle Augenstein. 2020. [A diagnostic study of explainability techniques for text classification](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3256–3274. Association for Computational Linguistics.
- David Bamman and Patrick J. Burns. 2020. [Latin BERT: A contextual language model for classical philology](#). *arXiv preprint arXiv:2009.10053*.
- Marc Brinner and Sina Zarrieß. 2023. [Model interpretability and rationale extraction by input mask optimization](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13722–13744. Association for Computational Linguistics.
- Patrick J. Burns, James A. Brofos, Kyle Li, Pramit Chaudhuri, and Joseph P. Dexter. 2021. [Profiling of intertextuality in Latin literature using word embeddings](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4900–4907. Association for Computational Linguistics.
- CCL. 2026a. [About the Project](#). Carolingian Canon Law Project home page, University of Kentucky. Accessed 2026-09-14.

733	CCL. 2026b. Submission Desks . Carolingian Canon	789
734	Law Project documentation page, University of Ken-	790
735	tucky. Accessed 2026-09-14.	791
736	Neil Coffee, Jean-Pierre Koenig, Shakthi Poornima,	792
737	Roelant Ossewaarde, Christopher Forstall, and Sarah	793
738	Jacobson. 2012. Intertextuality in the digital age .	794
739	<i>Transactions of the American Philological Association</i> ,	795
740	142(2):383–422.	796
741	Théotime de la Selle. 2026. From transcription to se-	797
742	mantic corpus analysis: Unsupervised learning of	
743	sentence representations for ancient languages . <i>arXiv</i>	798
744	<i>preprint arXiv:2607.24542</i> .	799
745	Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani,	800
746	Eric Lehman, Caiming Xiong, Richard Socher, and	801
747	Byron C. Wallace. 2020. ERASER: A benchmark to	802
748	evaluate rationalized NLP models . In <i>Proceedings</i>	803
749	<i>of the 58th Annual Meeting of the Association for</i>	
750	<i>Computational Linguistics</i> , pages 4443–4458, Online.	804
751	Association for Computational Linguistics.	805
752	Melodie H. Eichbauer. 2014. Review of the Carolin-	806
753	gian Canon Law Project , directed by Abigail Firey.	807
754	<i>Digital Philology: A Journal of Medieval Cultures</i> ,	808
755	3(1):165–168.	809
756	Kawin Ethayarajh. 2019. How contextual are contextu-	810
757	alized word representations? Comparing the geom-	811
758	etry of BERT, ELMo, and GPT-2 embeddings . In	812
759	<i>Proceedings of the 2019 Conference on Empirical</i>	
760	<i>Methods in Natural Language Processing and the</i>	813
761	<i>9th International Joint Conference on Natural Lan-</i>	814
762	<i>guage Processing (EMNLP-IJCNLP)</i> , pages 55–65.	815
763	Association for Computational Linguistics.	816
764	Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Ari-	817
765	vazhagan, and Wei Wang. 2022. Language-agnostic	818
766	BERT sentence embedding . In <i>Proceedings of the</i>	819
767	<i>60th Annual Meeting of the Association for Comput-</i>	820
768	<i>ational Linguistics (Volume 1: Long Papers)</i> , pages	
769	878–891. Association for Computational Linguistics.	821
770	Abigail Firey. 2009. The Carolingian Canon Law	822
771	Project . Collaboratory for Research in Computing	823
772	for the Humanities, University of Kentucky. Open-	824
773	access digital edition, ongoing. Accessed 2026-09-	825
774	14.	826
775	Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-	827
776	Yan Liu. 2019. Representation degeneration problem	
777	in training natural language generation models . In	828
778	<i>Proceedings of the 7th International Conference on</i>	829
779	<i>Learning Representations (ICLR)</i> .	830
780	Nathan Godey, Éric de la Clergerie, and Benoît Sagot.	831
781	2024. Anisotropy is inherent to self-attention in trans-	832
782	formers . In <i>Proceedings of the 18th Conference of</i>	833
783	<i>the European Chapter of the Association for Comput-</i>	834
784	<i>ational Linguistics (Volume 1: Long Papers)</i> , pages	
785	35–48. Association for Computational Linguistics.	835
786	Xinshuo Hu, Zifei Shan, Xinping Zhao, Zetian Sun,	836
787	Zhenyu Liu, Dongfang Li, Shaolin Ye, Xinyuan Wei,	837
788	Qian Chen, Baotian Hu, Haofen Wang, Jun Yu, and	838
	Min Zhang. 2025. KaLM-embedding: Superior train-	839
	ing data brings a stronger embedding model . <i>arXiv</i>	840
	<i>preprint arXiv:2501.01028</i> .	841
	Alon Jacovi and Yoav Goldberg. 2020. Towards faith-	842
	fully interpretable NLP systems: How should we	843
	define and evaluate faithfulness? In <i>Proceedings</i>	
	<i>of the 58th Annual Meeting of the Association for</i>	844
	<i>Computational Linguistics</i> , pages 4198–4205. Asso-	845
	ciation for Computational Linguistics.	
	Olga Kovaleva, Saurabh Kulshreshtha, Anna Rogers,	
	and Anna Rumshisky. 2021. BERT busters: Outlier	
	dimensions that disrupt transformers . In <i>Findings of</i>	
	<i>the Association for Computational Linguistics: ACL-</i>	
	<i>IJCNLP 2021</i> , pages 3392–3405. Association for	
	Computational Linguistics.	
	Nelson F. Liu, Matt Gardner, Yonatan Belinkov,	
	Matthew E. Peters, and Noah A. Smith. 2019. Lin-	
	guistic knowledge and transferability of contextual	
	representations . In <i>Proceedings of the 2019 Confer-</i>	
	<i>ence of the North American Chapter of the Associ-</i>	
	<i>ation for Computational Linguistics: Human Lan-</i>	
	<i>guage Technologies, Volume 1 (Long and Short Pa-</i>	
	<i>pers)</i> , pages 1073–1094. Association for Computa-	
	tional Linguistics.	
	Yifan Liu, Gelila Tilahun, Xinxiang Gao, Qianfeng Wen,	
	and Michael Gervers. 2025. A comparative study	
	of static and contextual embeddings for analyzing	
	semantic changes in medieval Latin charters . In <i>Pro-</i>	
	<i>ceedings of the First Workshop on Language Models</i>	
	<i>for Low-Resource Languages</i> , pages 182–192, Abu	
	Dhabi, United Arab Emirates. Association for Com-	
	putational Linguistics.	
	Anemily Machina and Robert Mercer. 2024. Anisotropy	
	is not inherent to transformers . In <i>Proceedings of</i>	
	<i>the 2024 Conference of the North American Chap-</i>	
	<i>ter of the Association for Computational Linguistics:</i>	
	<i>Human Language Technologies (Volume 1: Long</i>	
	<i>Papers)</i> , pages 4892–4907. Association for Compu-	
	tational Linguistics.	
	Enrique Manjavacas, Brian Long, and Mike Kestemont.	
	2019. On the feasibility of automated detection of	
	allusive text reuse . In <i>Proceedings of the 3rd Joint</i>	
	<i>SIGHUM Workshop on Computational Linguistics</i>	
	<i>for Cultural Heritage, Social Sciences, Humanities</i>	
	<i>and Literature</i> , pages 104–114. Association for Com-	
	putational Linguistics.	
	Leland McInnes, John Healy, and James Melville. 2018.	
	UMAP: Uniform manifold approximation and pro-	
	jection for dimension reduction . <i>arXiv preprint</i>	
	<i>arXiv:1802.03426</i> .	
	Jiaqi Mu and Pramod Viswanath. 2018. All-but-the-	
	Top: Simple and effective postprocessing for word	
	representations . In <i>Proceedings of the 6th Inter-</i>	
	<i>national Conference on Learning Representations</i>	
	<i>(ICLR)</i> .	
	Niklas Muennighoff, Nouamane Tazi, Loic Magne, and	
	Nils Reimers. 2023. MTEB: Massive text embedding	

benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2014–2037. Association for Computational Linguistics.

Jianmo Ni, Gustavo Hernandez Abrego, Noah Constant, Ji Ma, Keith Hall, Daniel Cer, and Yinfei Yang. 2022. [Sentence-T5: Scalable sentence encoders from pre-trained text-to-text models](#). In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1864–1874. Association for Computational Linguistics.

V. S. Raghu Parupudi. 2026. [Anisotropy decides cosine vs. rank metrics for text embeddings](#). *arXiv preprint arXiv:2606.29571*.

Sara Rajaei and Mohammad Taher Pilehvar. 2022. [An isotropy analysis in the multilingual BERT embedding space](#). In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1309–1316. Association for Computational Linguistics.

Anton Razhigaev, Matvey Mikhalechuk, Elizaveta Goncharova, Ivan Oseledets, Denis Dimitrov, and Andrey Kuznetsov. 2024. [The shape of learning: Anisotropy and intrinsic dimensions in transformer-based models](#). In *Findings of the Association for Computational Linguistics: EACL 2024*, pages 868–874. Association for Computational Linguistics.

Frederick Riemenschneider and Anette Frank. 2023a. [Exploring large language models for classical philology](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15181–15199. Association for Computational Linguistics.

Frederick Riemenschneider and Anette Frank. 2023b. [Graecia capta ferum victorem cepit. Detecting Latin allusions to Ancient Greek literature](#). In *Proceedings of the Ancient Language Processing Workshop*, pages 30–38, Varna, Bulgaria. INCOMA Ltd., Shoumen, Bulgaria.

Julian Schelb, Michael Wittweiler, Marie Revellio, Barbara Feichtinger, and Andreas Spitz. 2026. [Loci similes: A benchmark for extracting intertextualities in Latin literature](#). *arXiv preprint arXiv:2601.07533*.

Oscar Skea, Md Rifat Arefin, Dan Zhao, Niket Nikul Patel, Jalal Naghiyev, Yann LeCun, and Ravid Shwartz-Ziv. 2025. [Layer by layer: Uncovering hidden representations in language models](#). In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 55854–55875. PMLR.

David A. Smith, Ryan Cordell, and Elizabeth Maddock Dillon. 2013. [Infectious texts: Modeling text reuse in nineteenth-century newspapers](#). In *2013 IEEE International Conference on Big Data*, pages 86–94. IEEE.

Rachele Sprugnoli, Marco Passarotti, Flavio Massimiliano Cecchini, and Matteo Pellegrini. 2020. [Overview of the EvaLatin 2020 evaluation campaign](#). In *Proceedings of LT4HALA 2020 - 1st Workshop on Language Technologies for Historical and Ancient Languages*, pages 105–110. European Language Resources Association (ELRA).

Jianlin Su, Jiarun Cao, Weijie Liu, and Yangyiwen Ou. 2021. [Whitening sentence representations for better semantics and faster retrieval](#). *arXiv preprint arXiv:2103.15316*.

Mukund Sundararajan, Ankur Taly, and Qiqi Yan. 2017. [Axiomatic attribution for deep networks](#). In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *Proceedings of Machine Learning Research*, pages 3319–3328. PMLR.

Ian Tenney, Dipanjan Das, and Ellie Pavlick. 2019. [BERT rediscovers the classical NLP pipeline](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601. Association for Computational Linguistics.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. [BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models](#). In *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

William Timkey and Marten van Schijndel. 2021. [All bark and no bite: Rogue dimensions in transformer language models obscure representational quality](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 4527–4546. Association for Computational Linguistics.

Laurens van der Maaten and Geoffrey Hinton. 2008. [Visualizing data using t-SNE](#). *Journal of Machine Learning Research*, 9(86):2579–2605.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. [mT5: A massively multilingual pre-trained text-to-text transformer](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 483–498. Association for Computational Linguistics.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025. [Qwen3 embedding: Advancing text embedding and reranking through foundation models](#). *arXiv preprint arXiv:2506.05176*.

A Corpus Provenance and Derivation

This appendix records where the benchmark text comes from and how our files were derived from it, in the detail that Section 3.2 leaves out.

The project. The Carolingian Canon Law project (CCL; Firey, 2009) is an open-access digital edition founded in 2009; Eichbauer (2014) reviews it.¹ Its texts are early medieval canon-law manuscripts, most of them ninth-century and never edited to modern standards. Each manuscript is identified by its siglum (Place-Library-Shelfmark), and a witness is one manuscript's copy of one unit (one transcribed canon or excerpt).

Transcription. Transcription is manual, from manuscript images or, for some texts, eighteenth-century printings, with one exception: HTR has been attempted on one manuscript (Paris, BnF, lat. 1454; 142 of the 1,705 witnesses), and its transcription is still in proofreading. Before 2019 transcribers worked from purchased microfilm into word-processed files, which CCL contributors converted to TEI-P5 (the Text Encoding Initiative's XML standard for scholarly editions). Since 2019 they transcribe in the CCL Transcription Desk beside a IIIF image of the folio (a zoomable web-served scan of the manuscript leaf); an approved proofreader on the Proofreading Desk then checks the text against the image to the letter and adds TEI markup (CCL, 2026b). That proofreading stage belongs to the post-2019 workflow. The interface credits the transcriber and proofreader of each unit (CCL, 2026a).

Identifiers. The project has two identifier systems, both designed at its launch in 2009 by its director in consultation with a mathematician, a software developer and a second scholar of early medieval canon law. The first is a location identifier. Every manuscript is a unique object, so every unit has exactly one place in one manuscript, and the identifier records it as the manuscript siglum, the folium, and the unit's position on that folium. Our file names are these identifiers. The second is the source key. Many units in different manuscripts copy the same prescription issued by the same authority, and the key names that common source. It encodes the type of source (a church council, a papal decretal, a royal capitulary, a theologian, a penitential), the point of origin (a particular council, pope or author), the date of origin, and the item's reference in modern scholarship, such as a canon,

letter or register number. The key is designed so that a researcher can find the modern authoritative edition of the source where one exists.

Labels. CCL keys a unit to its original source, such as a fourth-century council or a papal decretal, once that source is identified, and our directories reuse that key as supplied. In a label, a four-letter code carries the type of source and its point of origin, followed by the year and the canon or chapter number, as in CTOU.567.16; some labels carry a trailing annotation, as in CTOU.567.16(15) or CARL.501?.18. Counting a directory as key-named when its label begins with a four-letter code, a period, a year and a period, whatever follows, 689 of the 840 directory labels are source keys. The other 151 are modern edition citations, the Apostolic Canons (Can.apost.N), references to other collections such as the Capitula Martini, biblical references, and a handful of unidentified sources; each unidentified source is a singleton directory and contributes no positive pairs.

Assignment and verification. Keys were assigned by the project's director, a specialist in early medieval canon law, or by graduate students working under the director's supervision, who had enough Latin to check a manuscript version against the modern authoritative version of the source. Verification, and occasional correction, runs through the project's parallel display, which sets side by side all units that share a key and shows how far their wording agrees. A misidentified unit stands out there as a text that cannot be a version of the others. This is verification by collation (see Limitations). The project supplied two label corrections during this study, and both are logged with the frozen benchmark. We assigned no label ourselves and changed only those two. What remains unidentified is large: across the several hundred manuscripts that contain the project's texts, thousands of units still carry no key, and they are the queries that Task B is meant to serve.

Derivation. We hold the witnesses as plain text derived from CCL TEI-P5 exports. The rule is one file per unit: the directory is the source key from the unit's scholSource division, which is stripped from the text so that no file carries its own label; the text is the manuscript rubric followed by the canon text, with inline markup flattened (editorial notes kept, deletions dropped) and layout whitespace collapsed; a unit without a source key

¹Founding year, pre-2019 workflow, proofreading standard, the one manuscript transcribed by handwritten text recognition (HTR), the design of the identifiers, the assignment and verification of keys, and the extent of unidentified material: unpublished documentation and correspondence supplied by the project, 2026.

receives no directory. The rule is scripted; checked against a current export of one manuscript (Paris, BnF, lat. 2123), it reproduces all 279 units present in both the export and the corpus, 100 percent after whitespace normalization with no character-level disagreement, and places every one of them in the same directory.

B Layer Diagnostics and Attribution Layer Selection

This appendix separates the most anisotropic layers, where geometry collapses, from the operational attribution layers, which are selected by training-set retrieval accuracy. Table 5 lists both for the three T5 encoders, together with the top-PC share and effective rank before and after ABTT. Figure 2 completes the layerwise view of Figure 1 by plotting Task A AUROC against depth for all six models; it is the view that shows the mT5-base collapse, a ranking failure that the cosine gap does not register because that gap rises in the middle layers.

Across the main-model layer grid, PC1 dominance and low effective rank are strongly associated with ABTT gains over baseline. These associations explain why ABTT helps, but they do not replace the train-only retrieval rule used for layer selection.

C Score Distributions at the Collapsed Retrieval Layer

Figure 3 restates the collapse and the repair as score distributions rather than as summary metrics, for the model whose collapse is easiest to read.

D Expanded Per-Layer Results for T5 Encoders

The main text compresses the headline retrieval results into Tables 2 and 3, which report one train-selected layer per model and method; Table 6 lists those layers. This appendix provides the expanded per-layer Task A duplicate-detection, Task B autonomous-routing, and Task B top- k ranking tables for LaTa, PhilTa, and mT5-base; Appendix F gives the same view for the other three models.

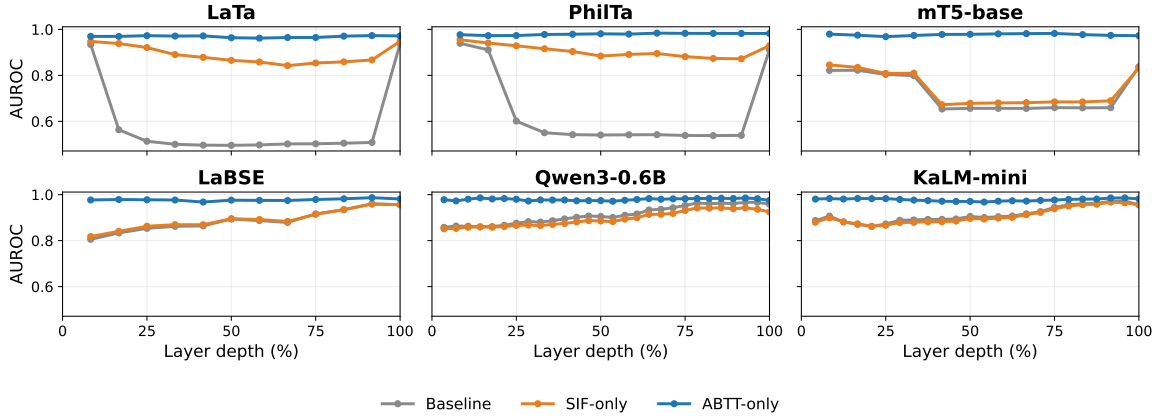


Figure 2: Task A AUROC against normalized layer depth, same panels and same series as Figure 1: gray is baseline mean pooling, blue is ABTT-only with D tuned on train, orange is SIF-only. In the top row, LaTa and PhilTa fall to chance in their middle layers and mT5-base loses more than half of its above-chance separation; for mT5-base this panel is the only view of the collapse, since its baseline cosine gap in Figure 1 rises over the same layers. The three non-T5 models in the bottom row improve close to monotonically with depth and never fall below 0.80. ABTT holds every model near its ceiling at every depth, so the repair does not depend on the collapse being present.

Model	Operational layer	Most anisotropic layer	PC1 before	PC1 after	Eff. rank before→after
LaTa	7	8	0.956	0.036	1.34→168.40
PhilTa	1	6	0.862	0.039	1.83→155.02
mT5-base	1	5	1.000	0.044	1.00→151.74

Table 5: Most anisotropic layers, by top-PC variance share, and operational attribution layers for the three T5 encoders. PC1 and effective-rank values compare baseline geometry with ABTT-D10 at the most anisotropic layer. Operational layers come from the train-only retrieval rule.

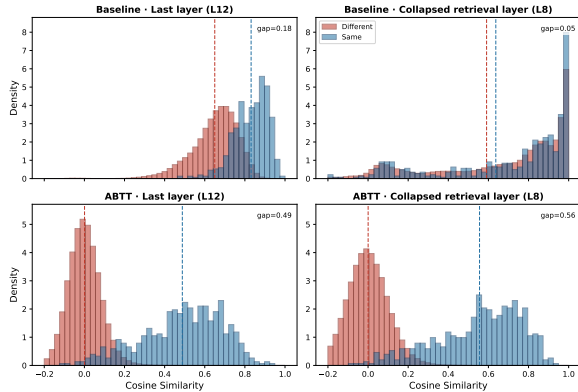


Figure 3: Cosine similarity distributions for PhilTa at the last layer and at its collapsed retrieval layer, before and after ABTT. Equivalent pairs share a directory (Same in the legend); non-equivalent pairs do not (Different). The collapsed retrieval layer is layer 8, the layer with the highest training-set AUROC within 30 to 70 percent of encoder depth (layers 4 to 8), so the panel understates the collapse rather than selecting its worst case; the same train-only rule that fixes every other choice in the paper picks it, and its test AUROC is 0.542, close to chance. Two other PhilTa layers carry other meanings and are not shown here: the most anisotropic layer under the top-PC rule of Appendix B is layer 6, and the lowest baseline test AUROC, 0.538, falls at layer 10. Before ABTT the two distributions sit almost on top of each other, and ABTT separates them.

Model	Task A layer				Task B layer			
	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	12	1	12	7	1	1	8	12
PhilTa	1	1	9	8	1	1	1	1
mT5-base	12	1	2	1	12	1	1	2
LaBSE	12	12	11	10	11	12	11	11
Qwen3-0.6B	26	22	2	2	28	27	5	5
KaLM-mini	23	23	1	2	23	22	3	1
LaTa (fine-tuned)	12	–	12	–	12	–	12	–
Qwen3-0.6B (fine-tuned)	28	–	27	–	28	–	27	–
KaLM-mini (fine-tuned)	24	–	24	–	24	–	24	–

Table 6: Train-selected layer behind every cell of Tables 2 and 3. For each model and post-processing setting, the Task A layer has the highest training-set AUROC and the Task B layer the highest training-set directory accuracy at rank 1; the test scores in those tables are read at these layers, so no layer is chosen on test. Layer 1 is the first transformer block. The fine-tuned rows below the rule have no SIF setting. Table 18 reads its pre-trained rows at the Base and ABTT layers above the rule and its fine-tuned rows at the layers below it.

Task A: Pairwise Duplicate Detection (main)					
Model	Layer	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
LaTa	1	0.934	0.970	0.286	0.566
	2	0.563	0.969	0.019	0.554
	3	0.513	0.973	-0.048	0.555
	4	0.500	0.971	-0.067	0.565
	5	0.496	0.972	-0.069	0.572
	6	0.496	0.964	-0.069	0.569
	7	0.498	0.962	-0.072	0.564
	8	0.502	0.965	-0.074	0.558
	9	0.503	0.965	-0.073	0.557
	10	0.505	0.971	-0.072	0.551
	11	0.508	0.973	-0.067	0.550
	12	0.938	0.971	0.237	0.525
PhilTa	1	0.939	0.977	0.231	0.574
	2	0.911	0.973	0.161	0.553
	3	0.601	0.973	0.085	0.549
	4	0.551	0.978	0.063	0.551
	5	0.542	0.979	0.053	0.551
	6	0.541	0.981	0.047	0.551
	7	0.542	0.979	0.046	0.554
	8	0.542	0.984	0.046	0.555
	9	0.539	0.983	0.041	0.540
	10	0.538	0.982	0.040	0.531
	11	0.539	0.982	0.041	0.511
	12	0.911	0.982	0.181	0.487
mT5-base	1	0.822	0.979	0.027	0.550
	2	0.822	0.975	0.026	0.536
	3	0.804	0.968	0.025	0.517
	4	0.799	0.974	0.040	0.502
	5	0.654	0.978	0.219	0.482
	6	0.657	0.978	0.238	0.483
	7	0.656	0.980	0.244	0.482
	8	0.656	0.982	0.249	0.471
	9	0.660	0.982	0.235	0.475
	10	0.659	0.977	0.226	0.459
	11	0.659	0.974	0.213	0.449
	12	0.838	0.972	0.081	0.442

Table 7: Per-layer Task A pairwise duplicate-detection metrics for the three T5 encoders (LaTa, PhilTa, mT5-base). baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. Both AUROC and cosine gap are computed over the test $n \times n$ cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set AUROC under ABTT. It is the layer behind the ABTT cell of Table 2 (listed in Table 6), and not always the layer with the highest test score in this table.

Task B: Autonomous Routing (main, file-level, τ -thresholded)							
Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
LaTa	1	0.630	0.910	0.916	0.907	0.738	0.909
	2	0.153	0.905	0.913	0.892	0.439	0.900
	3	0.256	0.882	0.842	0.950	0.477	0.908
	4	0.389	0.893	0.718	0.920	0.513	0.903
	5	0.374	0.892	0.715	0.913	0.502	0.900
	6	0.378	0.884	0.712	0.913	0.503	0.895
	7	0.372	0.875	0.715	0.916	0.501	0.890
	8	0.379	0.884	0.706	0.885	0.502	0.885
	9	0.366	0.886	0.724	0.873	0.501	0.881
	10	0.344	0.873	0.752	0.882	0.498	0.876
	11	0.310	0.867	0.786	0.901	0.490	0.880
	12	0.576	0.890	0.932	0.833	0.710	0.868
PhilTa	1	0.591	0.918	0.904	0.901	0.709	0.911
	2	0.479	0.895	0.867	0.885	0.625	0.892
	3	0.193	0.897	0.889	0.879	0.455	0.890
	4	0.148	0.886	0.923	0.873	0.439	0.881
	5	0.252	0.879	0.796	0.873	0.457	0.876
	6	0.338	0.905	0.700	0.817	0.474	0.872
	7	0.301	0.901	0.746	0.830	0.469	0.874
	8	0.262	0.875	0.808	0.885	0.467	0.879
	9	0.303	0.890	0.762	0.870	0.476	0.882
	10	0.299	0.908	0.780	0.793	0.480	0.865
	11	0.256	0.869	0.824	0.836	0.470	0.857
	12	0.338	0.880	0.898	0.876	0.549	0.879
mT5-base	1	0.135	0.877	0.981	0.947	0.453	0.903
	2	0.110	0.890	0.975	0.910	0.436	0.897
	3	0.161	0.794	0.926	0.941	0.449	0.850
	4	0.269	0.787	0.858	0.882	0.491	0.823
	5	0.422	0.746	0.659	0.864	0.512	0.790
	6	0.636	0.766	0.489	0.876	0.580	0.808
	7	0.594	0.793	0.517	0.851	0.565	0.815
	8	0.557	0.809	0.560	0.842	0.558	0.822
	9	0.364	0.841	0.746	0.814	0.508	0.831
	10	0.314	0.794	0.783	0.876	0.491	0.825
	11	0.297	0.763	0.780	0.854	0.479	0.797
	12	0.198	0.834	0.926	0.799	0.472	0.821

Table 8: Per-layer Task B autonomous routing accuracy for the three T5 encoders. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on train via F_1 -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbour’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 3 (listed in Table 6), and not always the layer with the highest test score in this table.

Task B: Top-K Ranking (main, single-seed)							
Model	Layer	Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
LaTa	1	0.721	0.880	0.630	0.910	0.916	0.907
	2	0.383	0.869	0.153	0.905	0.913	0.892
	3	0.350	0.882	0.256	0.882	0.842	0.950
	4	0.303	0.881	0.389	0.893	0.718	0.920
	5	0.297	0.875	0.374	0.892	0.715	0.913
	6	0.297	0.874	0.378	0.884	0.712	0.913
	7	0.302	0.868	0.372	0.875	0.715	0.916
	8	0.302	0.861	0.379	0.884	0.706	0.885
	9	0.309	0.858	0.366	0.886	0.724	0.873
	10	0.317	0.852	0.344	0.873	0.752	0.882
	11	0.331	0.857	0.310	0.867	0.786	0.901
	12	0.689	0.841	0.576	0.890	0.932	0.833
PhiTa	1	0.693	0.883	0.591	0.918	0.904	0.901
	2	0.603	0.869	0.479	0.895	0.867	0.885
	3	0.389	0.865	0.193	0.897	0.889	0.879
	4	0.369	0.859	0.148	0.886	0.923	0.873
	5	0.334	0.854	0.252	0.879	0.796	0.873
	6	0.310	0.845	0.338	0.905	0.700	0.817
	7	0.325	0.847	0.301	0.901	0.746	0.830
	8	0.350	0.857	0.262	0.875	0.808	0.885
	9	0.337	0.857	0.303	0.890	0.762	0.870
	10	0.344	0.833	0.299	0.908	0.780	0.793
	11	0.355	0.829	0.256	0.869	0.824	0.836
	12	0.528	0.854	0.338	0.880	0.898	0.876
mT5-base	1	0.451	0.885	0.135	0.877	0.981	0.947
	2	0.435	0.874	0.110	0.890	0.975	0.910
	3	0.445	0.830	0.161	0.794	0.926	0.941
	4	0.467	0.798	0.269	0.787	0.858	0.882
	5	0.352	0.756	0.422	0.746	0.659	0.864
	6	0.346	0.775	0.636	0.766	0.489	0.876
	7	0.352	0.775	0.594	0.793	0.517	0.851
	8	0.366	0.784	0.557	0.809	0.560	0.842
	9	0.394	0.787	0.364	0.841	0.746	0.814
	10	0.387	0.788	0.314	0.794	0.783	0.876
	11	0.378	0.765	0.297	0.763	0.780	0.854
	12	0.466	0.779	0.198	0.834	0.926	0.799

Table 9: Per-layer Task B top- k ranking metrics for the three T5 encoders. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. This table is computed on the single-seed split (no M -seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 17 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 3 (listed in Table 6), and not always the layer with the highest test score in this table.

E Last-Token vs. Mean Pooling

Decoder-only embedding models often use last-token pooling. We compare last-token pooling with mean pooling under the same ABTT pipeline, train/test split, and ABTT choices. Table 10 reports, for each model and pooling, the layer chosen on the train split by directory accuracy at rank 1, the rule of Table 3, with test-set Task B assignment accuracy, directory accuracy at rank 1, and cosine gap at that layer. Mean pooling performs better on assignment accuracy for every model in this experiment, including the decoder-only models.

F Per-Layer Results for Non-T5 Models

The main text focuses on LaTa, PhilTa, and mT5-base. This appendix reports the same per-layer view for LaBSE, Qwen3-0.6B, and KaLM-mini under the same comparison: baseline mean pooling versus ABTT with D tuned per layer on the train split.

Table 10: Mean versus last-token pooling under ABTT with D tuned per layer on the train split, for the five paper models whose last-token embeddings were extracted (mT5-base was not part of this run). For each model and pooling the layer is the one with the highest training-set directory accuracy at rank 1, the rule behind the ABTT cells of Table 3 (their layers are listed in Table 6); D is the number of principal components removed at that layer. The three score columns are test-set Task B assignment accuracy, Task B directory accuracy at rank 1, and Task A cosine gap at that layer.

Model	Pooling	Layer	D	Assignment acc.	Dir. acc. @1	Cosine gap
LaTa	Mean	8	10	0.885	0.861	0.558
	Last token	7	10	0.664	0.466	0.308
PhilTa	Mean	1	10	0.911	0.883	0.573
	Last token	11	10	0.589	0.406	0.205
LaBSE	Mean	11	10	0.908	0.885	0.610
	Last token	12	10	0.895	0.871	0.586
Qwen3-0.6B	Mean	5	10	0.915	0.894	0.545
	Last token	26	5	0.881	0.864	0.580
KaLM-mini	Mean	3	10	0.917	0.894	0.552
	Last token	24	10	0.875	0.847	0.483

Table 11: Per-layer Task A pairwise metrics for the non-T5 models (LaBSE, Qwen3-0.6B, KaLM-mini), under the same two-method comparison as the main paper. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. Both AUROC and cosine gap are computed over the test $n \times n$ cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set AUROC under ABTT. It is the layer behind the ABTT cell of Table 2 (listed in Table 6), and not always the layer with the highest test score in this table.

Task A: Pairwise Duplicate Detection (appendix models)					
Model	Layer	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
LaBSE	1	0.806	0.977	0.060	0.520
	2	0.833	0.979	0.037	0.531
	3	0.853	0.978	0.062	0.543
	4	0.862	0.977	0.073	0.525
	5	0.864	0.968	0.066	0.501
	6	0.894	0.976	0.078	0.532
	7	0.887	0.975	0.088	0.524
	8	0.879	0.975	0.080	0.522
	9	0.916	0.979	0.090	0.565
	10	0.934	0.982	0.103	0.586
	11	0.960	0.987	0.061	0.611
	12	0.956	0.981	0.183	0.592
Qwen3-0.6B	1	0.858	0.978	0.033	0.564
	2	0.863	0.973	0.024	0.553
	3	0.862	0.980	0.037	0.556
	4	0.860	0.986	0.036	0.547
	5	0.861	0.981	0.030	0.545
	6	0.867	0.983	0.035	0.545
	7	0.876	0.980	0.033	0.544
	8	0.882	0.972	0.037	0.532
	9	0.881	0.977	0.034	0.532
	10	0.887	0.976	0.031	0.529
	11	0.895	0.977	0.030	0.530
	12	0.901	0.973	0.027	0.530
	13	0.907	0.974	0.026	0.523
	14	0.905	0.974	0.024	0.519
	15	0.901	0.971	0.022	0.514
	16	0.910	0.975	0.026	0.509
	17	0.916	0.979	0.032	0.521
	18	0.933	0.983	0.032	0.537
	19	0.936	0.980	0.035	0.541
	20	0.943	0.982	0.041	0.559
	21	0.953	0.982	0.047	0.562
	22	0.963	0.983	0.044	0.583
	23	0.961	0.984	0.038	0.578
	24	0.963	0.984	0.032	0.576
	25	0.961	0.983	0.027	0.578
	26	0.966	0.985	0.024	0.568
	27	0.967	0.982	0.017	0.559
	28	0.958	0.975	0.098	0.556
KaLM-mini	1	0.887	0.981	0.063	0.568
	2	0.906	0.983	0.044	0.566
	3	0.881	0.981	0.062	0.551
	4	0.871	0.983	0.060	0.553
	5	0.861	0.983	0.062	0.541
	6	0.872	0.983	0.063	0.529
	7	0.888	0.979	0.072	0.524
	8	0.890	0.976	0.072	0.524
	9	0.892	0.974	0.066	0.507
	10	0.893	0.971	0.067	0.494
	11	0.894	0.970	0.061	0.475
	12	0.905	0.970	0.071	0.475
	13	0.900	0.968	0.066	0.467

(continued on next page)

Task A: Pairwise Duplicate Detection (appendix models)					
Model	Layer	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
	14	0.904	0.971	0.063	0.471
	15	0.907	0.973	0.067	0.468
	16	0.917	0.972	0.071	0.489
	17	0.926	0.975	0.075	0.501
	18	0.945	0.977	0.060	0.519
	19	0.956	0.979	0.061	0.531
	20	0.960	0.980	0.054	0.555
	21	0.962	0.981	0.043	0.555
	22	0.971	0.985	0.050	0.555
	23	0.972	0.986	0.056	0.552
	24	0.956	0.981	0.243	0.520

Table 12: Per-layer Task B autonomous routing accuracy for the non-T5 models. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on train via F_1 -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbour’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 3 (listed in Table 6), and not always the layer with the highest test score in this table.

Task B: Autonomous Routing (appendix models)							
Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
LaBSE	1	0.222	0.879	0.898	0.873	0.477	0.876
	2	0.381	0.875	0.811	0.827	0.543	0.857
	3	0.277	0.886	0.907	0.867	0.514	0.879
	4	0.372	0.897	0.885	0.836	0.565	0.874
	5	0.327	0.845	0.901	0.851	0.543	0.847
	6	0.548	0.860	0.882	0.851	0.674	0.857
	7	0.398	0.849	0.892	0.830	0.584	0.841
	8	0.385	0.869	0.870	0.743	0.568	0.822
	9	0.421	0.871	0.929	0.817	0.612	0.851
	10	0.606	0.892	0.892	0.805	0.713	0.859
	11	0.785	0.923	0.913	0.882	0.833	0.908
	12	0.729	0.914	0.938	0.864	0.808	0.895
Qwen3-0.6B	1	0.264	0.921	0.960	0.885	0.526	0.908
	2	0.437	0.869	0.814	0.916	0.579	0.887
	3	0.250	0.914	0.947	0.907	0.513	0.911
	4	0.411	0.936	0.848	0.842	0.576	0.901
	5	0.181	0.910	0.932	0.923	0.464	0.915
	6	0.422	0.935	0.830	0.796	0.576	0.882
	7	0.250	0.890	0.947	0.876	0.513	0.885
	8	0.409	0.931	0.895	0.799	0.592	0.881
	9	0.460	0.901	0.851	0.851	0.607	0.882
	10	0.533	0.892	0.836	0.864	0.647	0.881
	11	0.293	0.903	0.978	0.808	0.551	0.867
	12	0.407	0.912	0.938	0.817	0.607	0.876
	13	0.432	0.908	0.944	0.836	0.625	0.881
	14	0.456	0.907	0.926	0.811	0.633	0.871
	15	0.469	0.901	0.920	0.845	0.639	0.880
	16	0.639	0.897	0.830	0.808	0.711	0.864
	17	0.480	0.886	0.935	0.851	0.652	0.873
	18	0.553	0.882	0.944	0.839	0.700	0.866
	19	0.514	0.871	0.966	0.858	0.684	0.866
	20	0.654	0.865	0.898	0.892	0.746	0.875
	21	0.647	0.912	0.950	0.898	0.761	0.907
	22	0.731	0.888	0.938	0.916	0.809	0.899
	23	0.622	0.914	0.966	0.895	0.752	0.907

(continued on next page)

Task B: Autonomous Routing (appendix models)							
Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
	24	0.748	0.901	0.935	0.935	0.818	0.914
	25	0.574	0.893	0.975	0.923	0.725	0.904
	26	0.613	0.899	0.975	0.895	0.749	0.897
	27	0.793	0.877	0.833	0.923	0.808	0.894
	28	0.796	0.877	0.867	0.913	0.823	0.890
KaLM-mini	1	0.391	0.921	0.907	0.920	0.585	0.921
	2	0.493	0.916	0.848	0.879	0.627	0.902
	3	0.443	0.923	0.870	0.907	0.604	0.917
	4	0.308	0.931	0.935	0.895	0.544	0.917
	5	0.338	0.905	0.882	0.923	0.543	0.911
	6	0.256	0.895	0.910	0.920	0.502	0.904
	7	0.475	0.864	0.854	0.944	0.618	0.894
	8	0.467	0.862	0.870	0.913	0.619	0.881
	9	0.469	0.873	0.885	0.870	0.626	0.872
	10	0.449	0.875	0.885	0.870	0.613	0.873
	11	0.458	0.862	0.885	0.864	0.619	0.862
	12	0.507	0.862	0.895	0.904	0.653	0.878
	13	0.540	0.860	0.895	0.870	0.674	0.864
	14	0.593	0.836	0.845	0.885	0.688	0.854
	15	0.566	0.864	0.898	0.827	0.691	0.850
	16	0.550	0.849	0.935	0.830	0.695	0.841
	17	0.675	0.864	0.920	0.889	0.767	0.873
	18	0.697	0.875	0.960	0.898	0.796	0.883
	19	0.721	0.873	0.966	0.892	0.814	0.880
	20	0.705	0.791	0.966	0.913	0.803	0.837
	21	0.727	0.811	0.966	0.926	0.817	0.854
	22	0.850	0.879	0.916	0.923	0.875	0.895
	23	0.849	0.884	0.920	0.898	0.875	0.889
	24	0.787	0.892	0.944	0.867	0.846	0.882

Table 13: Per-layer Task B top- k ranking metrics for the non-T5 models. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. This table is computed on the single-seed split (no M -seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 17 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 3 (listed in Table 6), and not always the layer with the highest test score in this table.

Task B: Top-K Ranking (appendix models, single-seed)							
Model	Layer	Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
LaBSE	1	0.465	0.851	0.222	0.879	0.898	0.873
	2	0.494	0.830	0.381	0.875	0.811	0.827
	3	0.500	0.848	0.277	0.886	0.907	0.867
	4	0.542	0.841	0.372	0.897	0.885	0.836
	5	0.521	0.811	0.327	0.845	0.901	0.851
	6	0.640	0.831	0.548	0.860	0.882	0.851
	7	0.561	0.810	0.398	0.849	0.892	0.830
	8	0.544	0.779	0.385	0.869	0.870	0.743
	9	0.608	0.823	0.421	0.871	0.929	0.817
	10	0.693	0.832	0.606	0.892	0.892	0.805
	11	0.815	0.885	0.785	0.923	0.913	0.882
	12	0.791	0.868	0.729	0.914	0.938	0.864
Qwen3-0.6B	1	0.521	0.875	0.264	0.921	0.960	0.885
	2	0.558	0.871	0.437	0.869	0.814	0.916
	3	0.508	0.890	0.250	0.914	0.947	0.907
	4	0.559	0.874	0.411	0.936	0.848	0.842
	5	0.450	0.894	0.181	0.910	0.932	0.923
	6	0.551	0.858	0.422	0.935	0.830	0.796
	7	0.507	0.859	0.250	0.890	0.947	0.876

(continued on next page)

Task B: Top-K Ranking (appendix models, single-seed)							
Model	Layer	Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
	8	0.580	0.852	0.409	0.931	0.895	0.799
	9	0.587	0.852	0.460	0.901	0.851	0.851
	10	0.622	0.848	0.533	0.892	0.836	0.864
	11	0.548	0.830	0.293	0.903	0.978	0.808
	12	0.603	0.848	0.407	0.912	0.938	0.817
	13	0.618	0.851	0.432	0.908	0.944	0.836
	14	0.621	0.840	0.456	0.907	0.926	0.811
	15	0.627	0.848	0.469	0.901	0.920	0.845
	16	0.682	0.830	0.639	0.897	0.830	0.808
	17	0.641	0.840	0.480	0.886	0.935	0.851
	18	0.691	0.837	0.553	0.882	0.944	0.839
	19	0.676	0.837	0.514	0.871	0.966	0.858
	20	0.733	0.852	0.654	0.865	0.898	0.892
	21	0.752	0.881	0.647	0.912	0.950	0.898
	22	0.795	0.876	0.731	0.888	0.938	0.916
	23	0.742	0.886	0.622	0.914	0.966	0.895
	24	0.808	0.896	0.748	0.901	0.935	0.935
	25	0.718	0.886	0.574	0.893	0.975	0.923
	26	0.740	0.881	0.613	0.899	0.975	0.895
	27	0.787	0.876	0.793	0.877	0.833	0.923
	28	0.803	0.872	0.796	0.877	0.867	0.913
KaLM-mini	1	0.577	0.902	0.391	0.921	0.907	0.920
	2	0.607	0.879	0.493	0.916	0.848	0.879
	3	0.587	0.894	0.443	0.923	0.870	0.907
	4	0.538	0.887	0.308	0.931	0.935	0.895
	5	0.524	0.882	0.338	0.905	0.882	0.923
	6	0.490	0.876	0.256	0.895	0.910	0.920
	7	0.592	0.873	0.475	0.864	0.854	0.944
	8	0.597	0.860	0.467	0.862	0.870	0.913
	9	0.601	0.843	0.469	0.873	0.885	0.870
	10	0.584	0.844	0.449	0.875	0.885	0.870
	11	0.586	0.836	0.458	0.862	0.885	0.864
	12	0.622	0.850	0.507	0.862	0.895	0.904
	13	0.633	0.840	0.540	0.860	0.895	0.870
	14	0.636	0.829	0.593	0.836	0.845	0.885
	15	0.650	0.817	0.566	0.864	0.898	0.827
	16	0.676	0.819	0.550	0.849	0.935	0.830
	17	0.747	0.852	0.675	0.864	0.920	0.889
	18	0.777	0.865	0.697	0.875	0.960	0.898
	19	0.801	0.861	0.721	0.873	0.966	0.892
	20	0.789	0.817	0.705	0.791	0.966	0.913
	21	0.802	0.831	0.727	0.811	0.966	0.926
	22	0.857	0.875	0.850	0.879	0.916	0.923
	23	0.859	0.872	0.849	0.884	0.920	0.898
	24	0.833	0.858	0.787	0.892	0.944	0.867

G SIF and Whitening Post-Processing Sweeps

The main text uses ABTT-only as the paper-facing repair. This appendix reports SIF-only, SIF plus fixed- D ABTT, and SIF plus tuned- D ABTT, and gives the full account of the PCA whitening variant that Section 4.3 excludes. Table 14 first gives the five-seed cumulative top- K result discussed in the main text, with SIF plus ABTT at each model’s train-selected layer. The per-layer tables then keep one primary metric per task so the four-method grid stays readable.

Why whitening is excluded. PCA whitening (Su et al., 2021) leaves the different-directory cosine mean at or below 0.015, so the learned threshold degenerates: across all 100 whitening cells every file is routed to an existing directory, existing accuracy is 1.000, new accuracy never exceeds one file in 323, and assignment accuracy sits at the 62.4 percent class prior. That prior beats baseline in 60 of 100 cells while directory accuracy at rank 1 never exceeds 0.415, so the apparent gain is a thresholding artifact. SIF and whitening fall short of ABTT for different reasons. SIF acts before pooling, so it can only attenuate the frequent tokens that feed the dominant direction, and whatever survives that reweighting is still pooled into the passage vector; whitening acts after pooling but rescales all directions instead of deleting the top few, which flattens the same-directory margin along with the common component and so degenerates the routing threshold.

Model	Top-1	Top-2	Top-3	Top-4	Top-5
LaTa	88.9 $\pm$ 0.5	96.1 $\pm$ 0.5	97.4 $\pm$ 0.6	97.9 $\pm$ 0.3	98.2 $\pm$ 0.2
PhilTa	89.8 $\pm$ 0.8	95.8 $\pm$ 0.6	97.4 $\pm$ 0.3	97.7 $\pm$ 0.3	97.8 $\pm$ 0.4
mT5-base	89.3 $\pm$ 0.5	95.1 $\pm$ 0.2	96.7 $\pm$ 0.7	97.1 $\pm$ 0.4	97.2 $\pm$ 0.4
LaBSE	90.6 $\pm$ 1.0	95.5 $\pm$ 0.5	96.4 $\pm$ 0.5	97.2 $\pm$ 0.6	97.8 $\pm$ 0.5
Qwen3-0.6B	89.3 $\pm$ 0.7	95.7 $\pm$ 0.4	97.1 $\pm$ 0.3	97.5 $\pm$ 0.4	97.6 $\pm$ 0.4
KaLM-mini	89.3 $\pm$ 0.5	95.1 $\pm$ 0.4	96.6 $\pm$ 0.7	97.2 $\pm$ 0.7	97.6 $\pm$ 0.5

Table 14: Cumulative top- K accuracy for Task B under SIF+ABTT, as mean $\pm$ standard deviation over five query/reference reseeds at a fixed split. For each model the reported layer is chosen on the train split: the layer with the highest training-set directory accuracy at rank 1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 3, listed in Table 6. D is tuned per layer on the train split. This is not always the layer with the highest five-seed mean. Each row is the bold row of Table 17. Each cell is the percentage of test queries whose labelled directory appears within the top K options. Per-layer top- k rankings: Appendix Tables 9, 13, and 17.

Table 15: Per-layer Task A AUROC across the SIF-conditioned post-processing suite for all six models. `sif_only` replaces mean pooling with SIF-weighted pooling; `sif_abtt_fixed` adds ABTT with fixed $D=10$; `sif_abtt_optimal` tunes D per layer on the train split. Gap is omitted to keep the table narrow; the pure-ABTT comparison (not SIF-conditioned) is in Tables 7 and 11. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set AUROC under SIF+ABTT. It is the layer behind the SIF+ABTT cell of Table 2 (listed in Table 6), and not always the layer with the highest test score in this table.

Task A: SIF-suite AUROC (appendix)					
Model	Layer	AUROC (base)	AUROC (SIF)	AUROC (SIF+ABTT ₁₀)	AUROC (SIF+ABTT)
LaTa	1	0.934	0.948	0.972	0.976
	2	0.563	0.938	0.964	0.971
	3	0.513	0.921	0.967	0.966
	4	0.500	0.890	0.970	0.968
	5	0.496	0.878	0.970	0.970
	6	0.496	0.865	0.965	0.965
	7	0.498	0.858	0.966	0.966
	8	0.502	0.842	0.964	0.964
	9	0.503	0.854	0.964	0.964
	10	0.505	0.859	0.966	0.966
	11	0.508	0.867	0.965	0.965
	12	0.938	0.948	0.963	0.963
PhilTa	1	0.939	0.955	0.975	0.975
	2	0.911	0.941	0.973	0.973
	3	0.601	0.928	0.968	0.968
	4	0.551	0.916	0.972	0.972
	5	0.542	0.904	0.972	0.972
	6	0.541	0.884	0.981	0.981
	7	0.542	0.891	0.981	0.981
	8	0.542	0.895	0.982	0.982
	9	0.539	0.881	0.977	0.977
	10	0.538	0.874	0.977	0.977
	11	0.539	0.872	0.979	0.979
	12	0.911	0.929	0.969	0.969
mT5-base	1	0.822	0.845	0.974	0.974
	2	0.822	0.835	0.966	0.966
	3	0.804	0.808	0.958	0.958
	4	0.799	0.809	0.966	0.966
	5	0.654	0.672	0.974	0.974
	6	0.657	0.678	0.971	0.971
	7	0.656	0.680	0.967	0.967
	8	0.656	0.681	0.968	0.968
	9	0.660	0.685	0.967	0.967
	10	0.659	0.684	0.966	0.966
	11	0.659	0.689	0.966	0.966
	12	0.838	0.833	0.970	0.970
LaBSE	1	0.806	0.818	0.974	0.974
	2	0.833	0.840	0.975	0.975
	3	0.853	0.862	0.968	0.968
	4	0.862	0.869	0.966	0.966
	5	0.864	0.869	0.964	0.964
	6	0.894	0.895	0.965	0.965
	7	0.887	0.892	0.964	0.964
	8	0.879	0.883	0.968	0.968
	9	0.916	0.915	0.973	0.973
	10	0.934	0.934	0.976	0.976
	11	0.960	0.959	0.983	0.983
	12	0.956	0.956	0.980	0.984
Qwen3-0.6B	1	0.858	0.852	0.982	0.981
	2	0.863	0.853	0.972	0.972
	3	0.862	0.858	0.977	0.975
	4	0.860	0.860	0.976	0.977
	5	0.861	0.858	0.971	0.971
	6	0.867	0.861	0.974	0.974
	7	0.876	0.866	0.971	0.971

(continued on next page)

Task A: SIF-suite AUROC (appendix)					
Model	Layer	AUROC (base)	AUROC (SIF)	AUROC (SIF+ABTT ₁₀)	AUROC (SIF+ABTT)
	8	0.882	0.868	0.969	0.969
	9	0.881	0.866	0.968	0.968
	10	0.887	0.870	0.966	0.966
	11	0.895	0.875	0.964	0.964
	12	0.901	0.881	0.966	0.966
	13	0.907	0.887	0.967	0.967
	14	0.905	0.885	0.966	0.966
	15	0.901	0.883	0.968	0.968
	16	0.910	0.894	0.965	0.965
	17	0.916	0.899	0.967	0.974
	18	0.933	0.914	0.966	0.973
	19	0.936	0.915	0.966	0.976
	20	0.943	0.918	0.971	0.977
	21	0.953	0.931	0.971	0.971
	22	0.963	0.942	0.973	0.973
	23	0.961	0.941	0.973	0.973
	24	0.963	0.943	0.976	0.974
	25	0.961	0.938	0.978	0.975
	26	0.966	0.942	0.980	0.980
	27	0.967	0.936	0.978	0.978
	28	0.958	0.925	0.968	0.968
KaLM-mini	1	0.887	0.880	0.982	0.980
	2	0.906	0.899	0.981	0.981
	3	0.881	0.882	0.977	0.977
	4	0.871	0.873	0.974	0.974
	5	0.861	0.862	0.971	0.971
	6	0.872	0.867	0.967	0.967
	7	0.888	0.879	0.972	0.972
	8	0.890	0.880	0.973	0.975
	9	0.892	0.882	0.968	0.968
	10	0.893	0.882	0.964	0.964
	11	0.894	0.885	0.969	0.969
	12	0.905	0.896	0.966	0.966
	13	0.900	0.894	0.964	0.964
	14	0.904	0.898	0.970	0.970
	15	0.907	0.901	0.970	0.970
	16	0.917	0.913	0.967	0.967
	17	0.926	0.923	0.968	0.968
	18	0.945	0.939	0.977	0.977
	19	0.956	0.950	0.978	0.978
	20	0.960	0.956	0.978	0.978
	21	0.962	0.958	0.980	0.983
	22	0.971	0.967	0.983	0.984
	23	0.972	0.964	0.986	0.985
	24	0.956	0.956	0.976	0.976

Table 16: Per-layer Task B overall assignment accuracy across the SIF-conditioned suite for all six models. Restricted to overall routing accuracy (existing/new decomposition omitted) to keep the table compact; see Tables 8 and 12 for the pure-ABTT pairwise comparison with existing/new broken out. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under SIF+ABTT. It is the layer behind the SIF+ABTT cell of Table 3 (listed in Table 6), and not always the layer with the highest test score in this table.

Task B: SIF-suite routing (appendix)					
Model	Layer	Overall (base)	Overall (SIF)	Overall (SIF+ABTT ₁₀)	Overall (SIF+ABTT)
LaTa	1	0.738	0.832	0.920	0.899
	2	0.439	0.791	0.904	0.900
	3	0.477	0.720	0.910	0.907
	4	0.513	0.668	0.900	0.902
	5	0.502	0.648	0.906	0.906
	6	0.503	0.668	0.895	0.895

(continued on next page)

Task B: SIF-suite routing (appendix)					
Model	Layer	Overall (base)	Overall (SIF)	Overall (SIF+ABTT ₁₀)	Overall (SIF+ABTT)
	7	0.501	0.642	0.903	0.903
	8	0.502	0.614	0.892	0.892
	9	0.501	0.618	0.889	0.889
	10	0.498	0.655	0.890	0.890
	11	0.490	0.621	0.880	0.880
	12	0.710	0.804	0.902	0.902
PhilTa	1	0.709	0.790	0.914	0.914
	2	0.625	0.738	0.892	0.892
	3	0.455	0.713	0.892	0.892
	4	0.439	0.663	0.886	0.886
	5	0.457	0.626	0.899	0.899
	6	0.474	0.612	0.906	0.906
	7	0.469	0.625	0.906	0.906
	8	0.467	0.614	0.886	0.886
	9	0.476	0.590	0.895	0.895
	10	0.480	0.580	0.888	0.888
	11	0.470	0.565	0.886	0.886
	12	0.549	0.667	0.899	0.899
mT5-base	1	0.453	0.545	0.907	0.907
	2	0.436	0.537	0.903	0.903
	3	0.449	0.568	0.885	0.885
	4	0.491	0.488	0.857	0.857
	5	0.512	0.483	0.826	0.826
	6	0.580	0.558	0.817	0.817
	7	0.565	0.554	0.832	0.832
	8	0.558	0.548	0.834	0.834
	9	0.508	0.495	0.845	0.845
	10	0.491	0.495	0.832	0.832
	11	0.479	0.484	0.816	0.816
	12	0.472	0.541	0.858	0.858
LaBSE	1	0.477	0.521	0.886	0.886
	2	0.543	0.508	0.875	0.875
	3	0.514	0.577	0.860	0.860
	4	0.565	0.587	0.869	0.869
	5	0.543	0.598	0.857	0.857
	6	0.674	0.648	0.872	0.872
	7	0.584	0.634	0.858	0.858
	8	0.568	0.571	0.854	0.854
	9	0.612	0.613	0.859	0.859
	10	0.713	0.698	0.864	0.864
	11	0.833	0.807	0.896	0.896
	12	0.808	0.834	0.890	0.880
Qwen3-0.6B	1	0.526	0.566	0.914	0.909
	2	0.579	0.524	0.894	0.894
	3	0.513	0.537	0.904	0.901
	4	0.576	0.531	0.899	0.904
	5	0.464	0.537	0.908	0.908
	6	0.576	0.526	0.906	0.906
	7	0.513	0.573	0.911	0.911
	8	0.592	0.627	0.911	0.911
	9	0.607	0.576	0.910	0.910
	10	0.647	0.599	0.900	0.900
	11	0.551	0.606	0.893	0.893
	12	0.607	0.663	0.900	0.900
	13	0.625	0.675	0.897	0.897
	14	0.633	0.571	0.893	0.893
	15	0.639	0.600	0.893	0.893
	16	0.711	0.679	0.880	0.880
	17	0.652	0.686	0.900	0.876
	18	0.700	0.731	0.899	0.882
	19	0.684	0.700	0.886	0.850
	20	0.746	0.676	0.882	0.866

(continued on next page)

Task B: SIF-suite routing (appendix)					
Model	Layer	Overall (base)	Overall (SIF)	Overall (SIF+ABTT ₁₀)	Overall (SIF+ABTT)
KaLM-mini	21	0.761	0.739	0.892	0.892
	22	0.809	0.719	0.893	0.893
	23	0.752	0.759	0.893	0.893
	24	0.818	0.706	0.900	0.899
	25	0.725	0.748	0.897	0.893
	26	0.749	0.772	0.903	0.903
	27	0.808	0.760	0.899	0.899
	28	0.823	0.774	0.890	0.890
	1	0.585	0.604	0.906	0.908
	2	0.627	0.594	0.908	0.908
	3	0.604	0.576	0.907	0.907
	4	0.544	0.577	0.901	0.901
	5	0.543	0.533	0.897	0.897
	6	0.502	0.558	0.900	0.900
	7	0.618	0.593	0.901	0.901
	8	0.619	0.607	0.892	0.878
	9	0.626	0.663	0.876	0.876
	10	0.613	0.603	0.866	0.866
	11	0.619	0.627	0.874	0.874
	12	0.653	0.642	0.874	0.874
	13	0.674	0.666	0.864	0.864
	14	0.688	0.648	0.861	0.861
	15	0.691	0.698	0.866	0.866
	16	0.695	0.738	0.859	0.859
	17	0.767	0.763	0.882	0.882
	18	0.796	0.796	0.879	0.879
	19	0.814	0.804	0.878	0.878
	20	0.803	0.821	0.876	0.876
	21	0.817	0.832	0.874	0.876
	22	0.875	0.857	0.876	0.876
	23	0.875	0.862	0.880	0.881
	24	0.846	0.847	0.869	0.869

Table 17: Per-layer Task B top- k ranking accuracy across all six models, averaged over $M = 5$ query/reference reseeds of the train/test split (mean $\pm$ std). `sif_abtt_optimal` applies SIF weighting plus ABTT with D tuned per layer on the train split; the mseed sweep was not run for the ABTT-only variant of Tables 2 and 3, so we include this table to give a variance-aware view of the SIF-conditioned best variant. Overall assignment accuracy coincides with Acc@1 in the mseed pipeline, so we report the existing/new decomposition instead. Rows in bold mark the layer reported per model, the row printed in Table 14. For each model the reported layer is chosen on the train split: the layer with the highest training-set directory accuracy at rank 1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 3, listed in Table 6. D is tuned per layer on the train split. This is not always the layer with the highest five-seed mean.

Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT)							
Model	Layer	Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
LaTa	1	0.731 $\pm$ 0.010	0.889 $\pm$ 0.010	0.563 $\pm$ 0.007	0.844 $\pm$ 0.011	0.949 $\pm$ 0.011	0.948 $\pm$ 0.010
	2	0.429 $\pm$ 0.011	0.888 $\pm$ 0.007	0.049 $\pm$ 0.006	0.841 $\pm$ 0.007	0.920 $\pm$ 0.008	0.948 $\pm$ 0.007
	3	0.393 $\pm$ 0.011	0.891 $\pm$ 0.010	0.047 $\pm$ 0.006	0.847 $\pm$ 0.010	0.840 $\pm$ 0.020	0.947 $\pm$ 0.012
	4	0.331 $\pm$ 0.019	0.889 $\pm$ 0.009	0.048 $\pm$ 0.009	0.860 $\pm$ 0.009	0.696 $\pm$ 0.030	0.925 $\pm$ 0.013
	5	0.326 $\pm$ 0.021	0.891 $\pm$ 0.009	0.038 $\pm$ 0.005	0.858 $\pm$ 0.008	0.698 $\pm$ 0.028	0.934 $\pm$ 0.014
	6	0.322 $\pm$ 0.018	0.891 $\pm$ 0.009	0.038 $\pm$ 0.007	0.863 $\pm$ 0.010	0.688 $\pm$ 0.026	0.927 $\pm$ 0.013
	7	0.328 $\pm$ 0.020	0.896 $\pm$ 0.008	0.045 $\pm$ 0.007	0.848 $\pm$ 0.009	0.693 $\pm$ 0.028	0.957 $\pm$ 0.010
	8	0.325 $\pm$ 0.022	0.884 $\pm$ 0.008	0.046 $\pm$ 0.010	0.834 $\pm$ 0.008	0.685 $\pm$ 0.032	0.949 $\pm$ 0.011
	9	0.332 $\pm$ 0.019	0.886 $\pm$ 0.011	0.045 $\pm$ 0.010	0.848 $\pm$ 0.009	0.701 $\pm$ 0.024	0.935 $\pm$ 0.015
	10	0.346 $\pm$ 0.017	0.889 $\pm$ 0.004	0.045 $\pm$ 0.010	0.846 $\pm$ 0.008	0.735 $\pm$ 0.021	0.944 $\pm$ 0.012
	11	0.361 $\pm$ 0.018	0.882 $\pm$ 0.009	0.045 $\pm$ 0.009	0.862 $\pm$ 0.010	0.767 $\pm$ 0.029	0.907 $\pm$ 0.014
	12	0.707 $\pm$ 0.014	0.889 $\pm$ 0.005	0.511 $\pm$ 0.016	0.834 $\pm$ 0.005	0.961 $\pm$ 0.008	0.961 $\pm$ 0.010
PhilTa	1	0.705 $\pm$ 0.015	0.898 $\pm$ 0.008	0.531 $\pm$ 0.016	0.858 $\pm$ 0.010	0.931 $\pm$ 0.012	0.951 $\pm$ 0.011

(continued on next page)

Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT)							
Model	Layer	Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
	2	0.626 $\pm$ 0.019	0.888 $\pm$ 0.010	0.410 $\pm$ 0.017	0.866 $\pm$ 0.010	0.906 $\pm$ 0.018	0.917 $\pm$ 0.017
	3	0.440 $\pm$ 0.013	0.893 $\pm$ 0.009	0.071 $\pm$ 0.014	0.858 $\pm$ 0.010	0.917 $\pm$ 0.013	0.939 $\pm$ 0.011
	4	0.421 $\pm$ 0.016	0.891 $\pm$ 0.009	0.027 $\pm$ 0.007	0.858 $\pm$ 0.008	0.928 $\pm$ 0.015	0.935 $\pm$ 0.013
	5	0.375 $\pm$ 0.016	0.895 $\pm$ 0.010	0.042 $\pm$ 0.013	0.852 $\pm$ 0.010	0.804 $\pm$ 0.018	0.950 $\pm$ 0.014
	6	0.337 $\pm$ 0.015	0.897 $\pm$ 0.009	0.060 $\pm$ 0.012	0.861 $\pm$ 0.010	0.696 $\pm$ 0.026	0.944 $\pm$ 0.013
	7	0.361 $\pm$ 0.017	0.900 $\pm$ 0.009	0.056 $\pm$ 0.012	0.860 $\pm$ 0.008	0.755 $\pm$ 0.020	0.952 $\pm$ 0.013
	8	0.389 $\pm$ 0.016	0.891 $\pm$ 0.010	0.058 $\pm$ 0.014	0.859 $\pm$ 0.009	0.818 $\pm$ 0.018	0.932 $\pm$ 0.015
	9	0.371 $\pm$ 0.016	0.893 $\pm$ 0.008	0.066 $\pm$ 0.014	0.861 $\pm$ 0.009	0.766 $\pm$ 0.027	0.934 $\pm$ 0.016
	10	0.377 $\pm$ 0.014	0.892 $\pm$ 0.009	0.066 $\pm$ 0.014	0.855 $\pm$ 0.005	0.778 $\pm$ 0.023	0.940 $\pm$ 0.017
	11	0.394 $\pm$ 0.014	0.888 $\pm$ 0.008	0.059 $\pm$ 0.010	0.851 $\pm$ 0.010	0.827 $\pm$ 0.021	0.935 $\pm$ 0.013
	12	0.566 $\pm$ 0.013	0.897 $\pm$ 0.010	0.282 $\pm$ 0.014	0.863 $\pm$ 0.010	0.934 $\pm$ 0.007	0.941 $\pm$ 0.018
mT5-base	1	0.487 $\pm$ 0.012	0.899 $\pm$ 0.005	0.100 $\pm$ 0.009	0.856 $\pm$ 0.007	0.987 $\pm$ 0.007	0.954 $\pm$ 0.011
	2	0.476 $\pm$ 0.012	0.893 $\pm$ 0.005	0.083 $\pm$ 0.008	0.851 $\pm$ 0.012	0.984 $\pm$ 0.008	0.947 $\pm$ 0.013
	3	0.483 $\pm$ 0.014	0.883 $\pm$ 0.011	0.124 $\pm$ 0.009	0.817 $\pm$ 0.014	0.947 $\pm$ 0.008	0.970 $\pm$ 0.007
	4	0.500 $\pm$ 0.015	0.852 $\pm$ 0.007	0.206 $\pm$ 0.006	0.772 $\pm$ 0.010	0.879 $\pm$ 0.020	0.956 $\pm$ 0.010
	5	0.371 $\pm$ 0.014	0.825 $\pm$ 0.008	0.155 $\pm$ 0.006	0.724 $\pm$ 0.007	0.651 $\pm$ 0.023	0.956 $\pm$ 0.006
	6	0.361 $\pm$ 0.012	0.810 $\pm$ 0.005	0.263 $\pm$ 0.021	0.765 $\pm$ 0.012	0.488 $\pm$ 0.018	0.868 $\pm$ 0.016
	7	0.370 $\pm$ 0.010	0.827 $\pm$ 0.008	0.255 $\pm$ 0.014	0.760 $\pm$ 0.011	0.519 $\pm$ 0.017	0.914 $\pm$ 0.013
	8	0.382 $\pm$ 0.008	0.824 $\pm$ 0.012	0.244 $\pm$ 0.007	0.773 $\pm$ 0.012	0.561 $\pm$ 0.017	0.891 $\pm$ 0.016
	9	0.428 $\pm$ 0.014	0.838 $\pm$ 0.010	0.171 $\pm$ 0.008	0.783 $\pm$ 0.013	0.759 $\pm$ 0.024	0.909 $\pm$ 0.006
	10	0.421 $\pm$ 0.010	0.821 $\pm$ 0.017	0.140 $\pm$ 0.011	0.767 $\pm$ 0.015	0.783 $\pm$ 0.021	0.892 $\pm$ 0.019
	11	0.420 $\pm$ 0.015	0.803 $\pm$ 0.012	0.131 $\pm$ 0.012	0.721 $\pm$ 0.013	0.792 $\pm$ 0.016	0.910 $\pm$ 0.015
	12	0.507 $\pm$ 0.009	0.841 $\pm$ 0.009	0.169 $\pm$ 0.007	0.772 $\pm$ 0.016	0.942 $\pm$ 0.011	0.931 $\pm$ 0.011
LaBSE	1	0.502 $\pm$ 0.012	0.858 $\pm$ 0.007	0.170 $\pm$ 0.003	0.838 $\pm$ 0.007	0.929 $\pm$ 0.016	0.885 $\pm$ 0.009
	2	0.523 $\pm$ 0.011	0.860 $\pm$ 0.004	0.274 $\pm$ 0.005	0.817 $\pm$ 0.003	0.844 $\pm$ 0.022	0.915 $\pm$ 0.008
	3	0.535 $\pm$ 0.010	0.847 $\pm$ 0.006	0.218 $\pm$ 0.009	0.831 $\pm$ 0.009	0.945 $\pm$ 0.008	0.867 $\pm$ 0.007
	4	0.569 $\pm$ 0.008	0.850 $\pm$ 0.006	0.299 $\pm$ 0.004	0.837 $\pm$ 0.006	0.917 $\pm$ 0.014	0.866 $\pm$ 0.011
	5	0.551 $\pm$ 0.012	0.845 $\pm$ 0.005	0.255 $\pm$ 0.012	0.794 $\pm$ 0.005	0.933 $\pm$ 0.013	0.911 $\pm$ 0.010
	6	0.661 $\pm$ 0.011	0.859 $\pm$ 0.004	0.461 $\pm$ 0.019	0.817 $\pm$ 0.008	0.919 $\pm$ 0.018	0.914 $\pm$ 0.015
	7	0.596 $\pm$ 0.010	0.849 $\pm$ 0.004	0.333 $\pm$ 0.016	0.808 $\pm$ 0.006	0.935 $\pm$ 0.016	0.902 $\pm$ 0.010
	8	0.572 $\pm$ 0.012	0.847 $\pm$ 0.006	0.314 $\pm$ 0.019	0.802 $\pm$ 0.009	0.904 $\pm$ 0.016	0.905 $\pm$ 0.008
	9	0.631 $\pm$ 0.008	0.854 $\pm$ 0.007	0.381 $\pm$ 0.015	0.828 $\pm$ 0.002	0.954 $\pm$ 0.008	0.887 $\pm$ 0.012
	10	0.712 $\pm$ 0.016	0.858 $\pm$ 0.005	0.551 $\pm$ 0.019	0.836 $\pm$ 0.005	0.920 $\pm$ 0.015	0.887 $\pm$ 0.010
	11	0.820 $\pm$ 0.010	0.906 $\pm$ 0.010	0.731 $\pm$ 0.012	0.875 $\pm$ 0.013	0.935 $\pm$ 0.012	0.944 $\pm$ 0.017
	12	0.793 $\pm$ 0.015	0.884 $\pm$ 0.006	0.670 $\pm$ 0.019	0.850 $\pm$ 0.012	0.952 $\pm$ 0.013	0.927 $\pm$ 0.015
Qwen3-0.6B	1	0.550 $\pm$ 0.008	0.894 $\pm$ 0.005	0.223 $\pm$ 0.006	0.840 $\pm$ 0.010	0.972 $\pm$ 0.003	0.965 $\pm$ 0.011
	2	0.572 $\pm$ 0.015	0.882 $\pm$ 0.004	0.363 $\pm$ 0.014	0.828 $\pm$ 0.005	0.841 $\pm$ 0.032	0.952 $\pm$ 0.007
	3	0.541 $\pm$ 0.007	0.889 $\pm$ 0.006	0.216 $\pm$ 0.007	0.827 $\pm$ 0.008	0.962 $\pm$ 0.007	0.968 $\pm$ 0.011
	4	0.580 $\pm$ 0.012	0.891 $\pm$ 0.004	0.353 $\pm$ 0.008	0.845 $\pm$ 0.007	0.874 $\pm$ 0.028	0.950 $\pm$ 0.006
	5	0.487 $\pm$ 0.017	0.893 $\pm$ 0.007	0.133 $\pm$ 0.012	0.839 $\pm$ 0.011	0.944 $\pm$ 0.014	0.964 $\pm$ 0.009
	6	0.568 $\pm$ 0.012	0.897 $\pm$ 0.005	0.350 $\pm$ 0.010	0.857 $\pm$ 0.008	0.850 $\pm$ 0.023	0.948 $\pm$ 0.012
	7	0.541 $\pm$ 0.007	0.902 $\pm$ 0.006	0.211 $\pm$ 0.009	0.862 $\pm$ 0.008	0.967 $\pm$ 0.012	0.953 $\pm$ 0.012
	8	0.609 $\pm$ 0.009	0.903 $\pm$ 0.006	0.362 $\pm$ 0.013	0.861 $\pm$ 0.010	0.929 $\pm$ 0.017	0.956 $\pm$ 0.005
	9	0.612 $\pm$ 0.013	0.900 $\pm$ 0.004	0.393 $\pm$ 0.016	0.868 $\pm$ 0.008	0.894 $\pm$ 0.021	0.943 $\pm$ 0.006
	10	0.638 $\pm$ 0.013	0.890 $\pm$ 0.009	0.452 $\pm$ 0.009	0.847 $\pm$ 0.014	0.878 $\pm$ 0.022	0.944 $\pm$ 0.010
	11	0.578 $\pm$ 0.006	0.885 $\pm$ 0.005	0.257 $\pm$ 0.010	0.838 $\pm$ 0.014	0.991 $\pm$ 0.005	0.947 $\pm$ 0.012
	12	0.627 $\pm$ 0.009	0.892 $\pm$ 0.005	0.366 $\pm$ 0.016	0.844 $\pm$ 0.010	0.964 $\pm$ 0.013	0.955 $\pm$ 0.007
	13	0.639 $\pm$ 0.010	0.887 $\pm$ 0.006	0.383 $\pm$ 0.014	0.849 $\pm$ 0.011	0.970 $\pm$ 0.015	0.937 $\pm$ 0.011
	14	0.646 $\pm$ 0.009	0.885 $\pm$ 0.006	0.402 $\pm$ 0.017	0.852 $\pm$ 0.009	0.960 $\pm$ 0.016	0.927 $\pm$ 0.007
	15	0.648 $\pm$ 0.012	0.885 $\pm$ 0.007	0.410 $\pm$ 0.017	0.852 $\pm$ 0.010	0.954 $\pm$ 0.017	0.929 $\pm$ 0.018
	16	0.691 $\pm$ 0.016	0.873 $\pm$ 0.005	0.546 $\pm$ 0.017	0.837 $\pm$ 0.009	0.878 $\pm$ 0.017	0.920 $\pm$ 0.016
	17	0.658 $\pm$ 0.010	0.862 $\pm$ 0.006	0.420 $\pm$ 0.018	0.807 $\pm$ 0.012	0.966 $\pm$ 0.013	0.933 $\pm$ 0.013
	18	0.709 $\pm$ 0.012	0.868 $\pm$ 0.002	0.503 $\pm$ 0.018	0.833 $\pm$ 0.013	0.975 $\pm$ 0.006	0.914 $\pm$ 0.017
	19	0.691 $\pm$ 0.010	0.874 $\pm$ 0.002	0.462 $\pm$ 0.018	0.860 $\pm$ 0.007	0.986 $\pm$ 0.006	0.893 $\pm$ 0.012
	20	0.743 $\pm$ 0.020	0.860 $\pm$ 0.004	0.588 $\pm$ 0.023	0.797 $\pm$ 0.011	0.943 $\pm$ 0.018	0.941 $\pm$ 0.010
	21	0.758 $\pm$ 0.017	0.879 $\pm$ 0.006	0.585 $\pm$ 0.021	0.836 $\pm$ 0.008	0.981 $\pm$ 0.010	0.935 $\pm$ 0.010
	22	0.799 $\pm$ 0.014	0.882 $\pm$ 0.006	0.671 $\pm$ 0.019	0.855 $\pm$ 0.002	0.965 $\pm$ 0.006	0.918 $\pm$ 0.015
	23	0.757 $\pm$ 0.012	0.887 $\pm$ 0.006	0.577 $\pm$ 0.014	0.857 $\pm$ 0.007	0.989 $\pm$ 0.004	0.927 $\pm$ 0.011
	24	0.810 $\pm$ 0.014	0.893 $\pm$ 0.005	0.695 $\pm$ 0.021	0.861 $\pm$ 0.004	0.959 $\pm$ 0.003	0.935 $\pm$ 0.011
	25	0.732 $\pm$ 0.011	0.890 $\pm$ 0.009	0.527 $\pm$ 0.016	0.866 $\pm$ 0.006	0.997 $\pm$ 0.003	0.920 $\pm$ 0.014
	26	0.749 $\pm$ 0.013	0.895 $\pm$ 0.008	0.562 $\pm$ 0.017	0.858 $\pm$ 0.004	0.991 $\pm$ 0.003	0.943 $\pm$ 0.015
	27	0.812 $\pm$ 0.004	0.881 $\pm$ 0.005	0.741 $\pm$ 0.008	0.824 $\pm$ 0.006	0.903 $\pm$ 0.011	0.954 $\pm$ 0.011

(continued on next page)

Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT)							
Model	Layer	Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
	28	0.824 $\pm$ 0.010	0.871 $\pm$ 0.006	0.736 $\pm$ 0.015	0.802 $\pm$ 0.010	0.939 $\pm$ 0.005	0.961 $\pm$ 0.007
KaLM-mini	1	0.596 $\pm$ 0.012	0.893 $\pm$ 0.005	0.344 $\pm$ 0.003	0.846 $\pm$ 0.005	0.922 $\pm$ 0.022	0.953 $\pm$ 0.006
	2	0.620 $\pm$ 0.018	0.893 $\pm$ 0.003	0.422 $\pm$ 0.014	0.845 $\pm$ 0.005	0.876 $\pm$ 0.023	0.955 $\pm$ 0.008
	3	0.605 $\pm$ 0.011	0.893 $\pm$ 0.004	0.379 $\pm$ 0.008	0.843 $\pm$ 0.000	0.896 $\pm$ 0.018	0.959 $\pm$ 0.010
	4	0.562 $\pm$ 0.015	0.889 $\pm$ 0.005	0.265 $\pm$ 0.009	0.842 $\pm$ 0.008	0.945 $\pm$ 0.016	0.949 $\pm$ 0.008
	5	0.550 $\pm$ 0.013	0.890 $\pm$ 0.004	0.270 $\pm$ 0.005	0.841 $\pm$ 0.004	0.910 $\pm$ 0.019	0.952 $\pm$ 0.009
	6	0.522 $\pm$ 0.016	0.893 $\pm$ 0.004	0.203 $\pm$ 0.012	0.850 $\pm$ 0.005	0.934 $\pm$ 0.014	0.949 $\pm$ 0.014
	7	0.604 $\pm$ 0.018	0.893 $\pm$ 0.003	0.393 $\pm$ 0.018	0.840 $\pm$ 0.008	0.876 $\pm$ 0.023	0.962 $\pm$ 0.008
	8	0.607 $\pm$ 0.014	0.866 $\pm$ 0.003	0.388 $\pm$ 0.011	0.798 $\pm$ 0.008	0.891 $\pm$ 0.017	0.954 $\pm$ 0.007
	9	0.614 $\pm$ 0.011	0.874 $\pm$ 0.005	0.387 $\pm$ 0.008	0.821 $\pm$ 0.008	0.908 $\pm$ 0.013	0.943 $\pm$ 0.008
	10	0.602 $\pm$ 0.013	0.859 $\pm$ 0.008	0.360 $\pm$ 0.005	0.774 $\pm$ 0.010	0.916 $\pm$ 0.017	0.969 $\pm$ 0.010
	11	0.601 $\pm$ 0.012	0.864 $\pm$ 0.008	0.361 $\pm$ 0.007	0.792 $\pm$ 0.010	0.911 $\pm$ 0.020	0.956 $\pm$ 0.008
	12	0.640 $\pm$ 0.010	0.868 $\pm$ 0.006	0.416 $\pm$ 0.006	0.801 $\pm$ 0.007	0.929 $\pm$ 0.019	0.954 $\pm$ 0.006
	13	0.645 $\pm$ 0.011	0.862 $\pm$ 0.005	0.436 $\pm$ 0.005	0.793 $\pm$ 0.004	0.915 $\pm$ 0.018	0.951 $\pm$ 0.012
	14	0.654 $\pm$ 0.010	0.859 $\pm$ 0.006	0.479 $\pm$ 0.004	0.795 $\pm$ 0.008	0.880 $\pm$ 0.017	0.941 $\pm$ 0.011
	15	0.661 $\pm$ 0.009	0.870 $\pm$ 0.006	0.460 $\pm$ 0.009	0.814 $\pm$ 0.009	0.921 $\pm$ 0.014	0.941 $\pm$ 0.006
	16	0.685 $\pm$ 0.008	0.870 $\pm$ 0.007	0.477 $\pm$ 0.009	0.824 $\pm$ 0.008	0.953 $\pm$ 0.015	0.930 $\pm$ 0.008
	17	0.757 $\pm$ 0.006	0.886 $\pm$ 0.010	0.612 $\pm$ 0.016	0.841 $\pm$ 0.005	0.944 $\pm$ 0.010	0.943 $\pm$ 0.016
	18	0.781 $\pm$ 0.008	0.880 $\pm$ 0.008	0.636 $\pm$ 0.013	0.840 $\pm$ 0.002	0.967 $\pm$ 0.007	0.931 $\pm$ 0.016
	19	0.805 $\pm$ 0.012	0.877 $\pm$ 0.008	0.673 $\pm$ 0.020	0.831 $\pm$ 0.007	0.976 $\pm$ 0.003	0.935 $\pm$ 0.011
	20	0.794 $\pm$ 0.010	0.878 $\pm$ 0.006	0.650 $\pm$ 0.018	0.831 $\pm$ 0.003	0.979 $\pm$ 0.004	0.937 $\pm$ 0.011
	21	0.805 $\pm$ 0.010	0.871 $\pm$ 0.007	0.673 $\pm$ 0.016	0.831 $\pm$ 0.004	0.975 $\pm$ 0.006	0.924 $\pm$ 0.012
	22	0.871 $\pm$ 0.006	0.873 $\pm$ 0.006	0.811 $\pm$ 0.005	0.813 $\pm$ 0.007	0.949 $\pm$ 0.006	0.951 $\pm$ 0.007
	23	0.873 $\pm$ 0.007	0.885 $\pm$ 0.005	0.811 $\pm$ 0.006	0.835 $\pm$ 0.007	0.953 $\pm$ 0.007	0.950 $\pm$ 0.006
	24	0.848 $\pm$ 0.009	0.856 $\pm$ 0.004	0.749 $\pm$ 0.017	0.807 $\pm$ 0.006	0.975 $\pm$ 0.007	0.918 $\pm$ 0.009

H Reference Systems: the Fine-Tuning Ceilings

Tables 2 and 3 carry three reference rows below the rule, one supervised fine-tuning ceiling per fine-tuned model. Table 18 gives all three with the pre-trained rows they are compared against. Every model we fine-tuned is reported, on whichever side of its own zero-shot row it lands.

Three encoders are fine-tuned under one recipe, LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning only the Latin-pretrained model would leave the ceiling confounded with Latin pre-training, so the second model is a multilingual decoder-only encoder with no Latin-specific pre-training, Qwen3-0.6B. The third answers the remaining question, whether supervision can clear the best cell the correction produces anywhere in the paper: KaLM-mini holds it, at 91.7 assignment accuracy and at 89.4 directory accuracy at rank 1 tied with Qwen3-0.6B. All three are trained with symmetric InfoNCE over positive pairs and in-batch negatives (no two pairs from one directory in a batch), temperature 0.05, AdamW at learning rate 2×10^{-5} with weight decay 0.01, linear warmup over the first 10 percent of steps then linear decay, gradient clipping at 1.0, 16 pairs per batch, up to 8 epochs with early stopping after 3 epochs without dev improvement, and seed 42. All three are mean-pooled with the same token filter as their zero-shot rows, so the fine-tuned vectors live in the space those rows are measured in; Qwen3-0.6B and KaLM-mini add gradient checkpointing, which recomputes activations rather than storing them and leaves the gradients unchanged. The dev slice holds out 28 of the 190 train directories with at least two files (71 files), the same directories for every model, so each trains on 499 of the 565 positive train pairs, and the checkpoint is chosen by dev directory accuracy at rank 1 with dev AUROC as the tie-break. Accuracy differences at or below one dev file are treated as ties, because one file out of 71 is the pool’s whole resolution; without that, a model whose pre-trained encoder already routes all 71 dev files can never select a trained checkpoint. A 71-file pool is the weak link in this protocol: one borderline file flipped between two otherwise identical Qwen3-0.6B runs at a training loss of 6×10^{-4} , which is GPU non-determinism rather than a difference in the model, and it is worth 1.4 points of the selection criterion. LaTa selects epoch 7, the last epoch run before early stopping fired, and its ABTT sweep selects

$D = 10$, the top of the grid, at every layer, so the tuned and fixed ABTT variants coincide there. Qwen3-0.6B selects epoch 3 of 6 run, and its sweep is a genuine one, selecting the top of the grid at 12 of 28 layers and $D = 2$ at the layer its Task B row reports. KaLM-mini selects epoch 1 of 4 run, the only model of the three whose pre-trained encoder leaves the dev pool room to improve, and its sweep selects the top of the grid at 19 of 24 layers including the layer its Task B row reports. All three rows are ceilings at this training budget rather than asymptotes: training loss reaches 4×10^{-5} on Qwen3-0.6B by epoch 4 and 3×10^{-5} on KaLM-mini, so the 499 pairs are memorised and more epochs buy nothing.

System	Task A		Task B	
	AUROC	Cosine gap	Assign. acc.	Dir. acc.@1
LaTa (pre-trained)	0.938	0.237	73.8	72.1
LaTa (pre-trained) + ABTT	0.971	0.525	88.5	86.1
LaTa (fine-tuned)	0.984	0.387	83.4	81.6
LaTa (fine-tuned) + ABTT	0.970	0.548	87.8	85.2
Qwen3-0.6B (pre-trained)	0.966	0.024	82.3	80.3
Qwen3-0.6B (pre-trained) + ABTT	0.973	0.553	91.5	89.4
Qwen3-0.6B (fine-tuned)	0.996	0.684	91.4	90.8
Qwen3-0.6B (fine-tuned) + ABTT	0.994	0.716	92.1	91.4
KaLM-mini (pre-trained)	0.972	0.056	87.5	85.9
KaLM-mini (pre-trained) + ABTT	0.981	0.568	91.7	89.4
KaLM-mini (fine-tuned)	0.997	0.636	92.4	91.7
KaLM-mini (fine-tuned) + ABTT	0.994	0.651	93.4	92.5

Table 18: Supervised fine-tuning reference ceiling on LaTa, Qwen3-0.6B and KaLM-mini. The fine-tuned encoders are trained contrastively on 499 of the 565 positive train pairs (in-batch negatives, symmetric InfoNCE), with a dev slice carved out of train by directory for model selection; the test split is untouched until this evaluation. Task A columns are test AUROC and cosine gap at the layer chosen by train AUROC; Task B columns are test assignment accuracy and directory accuracy at rank 1, in percent, at the layer chosen by train directory accuracy, with τ learned on train. The selected layers are listed in Table 6. ABTT rows for LaTa sweep D per layer on train and select $D = 10$ everywhere, which is the top of the grid $\{1, 2, 3, 5, 7, 10\}$, so the tuned and fixed variants coincide. The selected checkpoint for LaTa is epoch 7, the terminal epoch of the sweep under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for Qwen3-0.6B sweep D per layer on train over the grid $\{1, 2, 3, 5, 7, 10\}$; 12 of 28 layer rows select the top of the grid. The selected checkpoint for Qwen3-0.6B is epoch 3 of 6 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for KaLM-mini sweep D per layer on train over the grid $\{1, 2, 3, 5, 7, 10\}$; 19 of 24 layer rows select the top of the grid. The selected checkpoint for KaLM-mini is epoch 1 of 4 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. This is a reference ceiling, not a proposed method: it consumes supervision the zero-shot pipeline never sees.

I Secondary Attribution Metrics and Sweeps

Ranking conventions. Two ranking conventions hold for every curve metric and every method, variant and control alike. Tokens are ranked by attribution magnitude $|a|$ rather than by signed importance, because the question is which tokens carry the pair cosine and both methods produce two-signed scores. Ties are broken by token position, which matters most for MaRC, whose many exactly-zero scores are then ordered by reading order. Absolute values are therefore not comparable with a signed-ranking implementation, and Kendall τ_b in Table 19 is the tie-corrected companion to ρ_{LOO} under the second convention.

Secondary metrics. Figure 4 plots the rank-faithfulness column of Table 4. Table 19 carries the metrics the main table does not: tie-corrected Kendall τ_b , chance-corrected insertion faithfulness, and the three ERASER-style rationale metrics at the global threshold settings. Sufficiency keeps the top-ranked query tokens and measures how much of the full pair cosine remains, comprehensiveness removes those tokens and measures the score drop, and MinFrac reports the smallest retained query-token fraction needed to recover a fixed share of the full cosine. Five computed quantities are printed in neither table: the raw deletion and insertion areas, which appear here only as the chance-corrected gaps; AOPC-Sufficiency and AOPC-Comprehensiveness, whose per-threshold components are in the sweeps below; and the per-cell shuffled-attribution gaps, which the main text reports as a validity statement rather than as a result column. The rest of this appendix expands those rationale metrics across Sufficiency and Comprehensiveness thresholds of 10%, 25%, and 50%, and MinFrac recovery thresholds of 0.70, 0.80, 0.90, and 0.95. We use one global midpoint setting rather than selecting thresholds per model or per method. Table 20 gives the sweep for integrated gradients and retrieval-adapted MaRC, and Table 21 gives it for the supplemental attribution methods. Every table in this appendix uses the same representation-level erasure operator as the main table, so the cells are comparable across them.

J Qualitative Attribution Examples

This appendix keeps the example-level views for one PhilTa retrieval pair. Figure 5 gives the

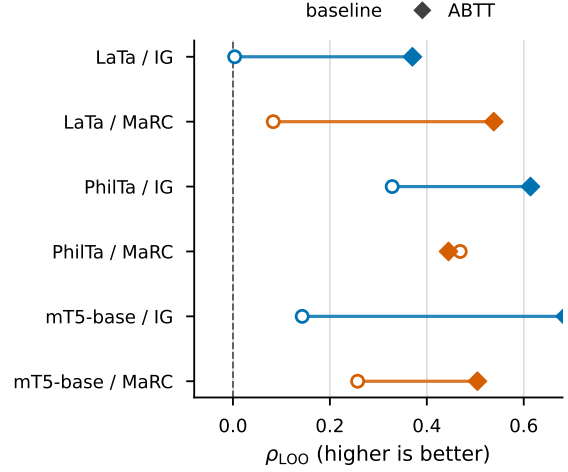


Figure 4: Leave-one-out rank correlation ρ_{LOO} at the predeclared operational attribution layers, for integrated gradients (IG) and retrieval-adapted MaRC. Each line connects the baseline and ABTT variants of one model-method cell. ABTT raises ρ_{LOO} in five of the six cells; the sixth, PhilTa MaRC, moves from 0.469 to 0.445, a tie within two standard errors (Table 4). Secondary metrics: Table 19.

integrated-gradient pair matrix before and after ABTT. The attention map in Figure 6 is context only, since ABTT is applied after encoding. The retrieval-adapted MaRC masks in Figure 7 show the learned-mask view for the same pair.

K Cluster Geometry

Figures 8 and 9 give the t-SNE and UMAP panels for LaTa, PhilTa, and mT5-base, and Figures 10 and 11 extend them to LaBSE, Qwen3-0.6B, and KaLM-mini. All four draw all 1,705 witnesses, color the same six largest directories, and compare baseline mean pooling with ABTT-only at $D = 10$; each model sits at the layer chosen on the train split by directory accuracy at rank 1 under ABTT, the Task B ABTT layer of Table 6.

Model	Method	τ_{LOO}		InsAUC gap		Suff@25%		Comp@25%		MinFrac@0.80	
		base	ABTT	base	ABTT	base	ABTT	base	ABTT	base	ABTT
LaTa	IG	0.007	0.262	0.284	0.061	0.990	0.804	1.242	0.177	0.111	0.272
	MaRC	0.068	0.400	0.088	0.138	0.731	0.917	0.781	0.280	0.184	0.177
PhilTa	IG	0.234	0.462	0.004	0.245	0.814	0.971	0.113	0.522	0.240	0.136
	MaRC	0.334	0.318	0.063	0.193	0.888	0.878	0.093	0.363	0.149	0.259
mT5-base	IG	0.101	0.515	-0.024	0.399	0.933	1.126	0.012	0.642	0.086	0.178
	MaRC	0.181	0.359	0.003	0.327	0.960	1.013	0.014	0.582	0.031	0.246

Table 19: Secondary attribution metrics, on the same pairs, the same layers and the same erasure operator as Table 4. Boldface marks the better variant within a pair; higher is better everywhere except MinFrac. None of these columns is in the main table, and each is out for its own reason. Kendall τ_b agrees with ρ_{LOO} in 5/6 cells, but it is the tie-corrected twin of the same statistic rather than a second witness. Chance-corrected insertion faithfulness favours ABTT in 5/6 cells, and we do not report it in the main table because in one of the twelve cells, a baseline cell, the real attribution does not beat a permutation of its own scores, so the measurement does not meet the validity bar we set for a headline column. The threshold-based ERASER metrics are reported for completeness: their baseline-versus-ABTT verdict depends on the threshold and on the erasure operator, which is why the main table uses threshold-free, chance-corrected metrics instead. The InsAUC columns average 195 to 199 ABTT pairs against the baseline’s 199 to 200, for the same small-denominator reason. Full threshold sweeps are in Tables 20 and 21.

Model	Method	$\rho_{\text{LOO}} \uparrow$		Suff@10% $\uparrow$		Suff@25% $\uparrow$		Suff@50% $\uparrow$		Comp@10% $\uparrow$		Comp@25% $\uparrow$		Comp@50% $\uparrow$		MinFrac@0.70 $\downarrow$		MinFrac@0.80 $\downarrow$		MinFrac@0.90 $\downarrow$		MinFrac@0.95 $\downarrow$	
		base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt
LaTa	IG	0.003	0.370	0.949	0.556	0.990	0.804	1.014	0.969	1.144	0.069	1.242	0.177	1.203	0.354	0.082	0.196	0.111	0.272	0.159	0.388	0.197	0.475
	MaRC	0.083	0.538	0.696	0.797	0.731	0.917	0.807	1.004	0.487	0.112	0.781	0.280	0.889	0.556	0.139	0.101	0.184	0.177	0.270	0.328	0.317	0.430
PhilTa	IG	0.329	0.614	0.637	0.844	0.814	0.971	0.937	1.038	0.042	0.288	0.113	0.522	0.231	0.746	0.156	0.084	0.240	0.136	0.411	0.221	0.548	0.290
	MaRC	0.469	0.445	0.786	0.691	0.888	0.878	0.971	1.001	0.032	0.169	0.093	0.363	0.263	0.625	0.076	0.199	0.149	0.259	0.299	0.350	0.435	0.406
mT5-base	IG	0.143	0.686	0.839	0.903	0.933	1.126	0.976	1.210	0.004	0.312	0.012	0.642	0.038	1.005	0.045	0.126	0.086	0.178	0.184	0.252	0.323	0.304
	MaRC	0.257	0.504	0.918	0.791	0.960	1.013	0.989	1.121	0.004	0.274	0.014	0.582	0.042	0.902	0.016	0.175	0.031	0.246	0.085	0.347	0.199	0.412

Table 20: Appendix attribution sweep for the two paper-facing attribution views, IG and retrieval-adapted MARC, across the three main-paper models. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at {10%, 25%, 50%}; MinFrac is the minimum retained query-token fraction needed to recover $\tau \in \{0.70, 0.80, 0.90, 0.95\}$ of the full cosine (lower is better). Ratio metrics require $|S_v(q_{\text{full}}, c)| \geq 0.05$; ρ_{LOO} excludes $|\Delta| < 10^{-6}$. The completeness report contains no missing metric cells.

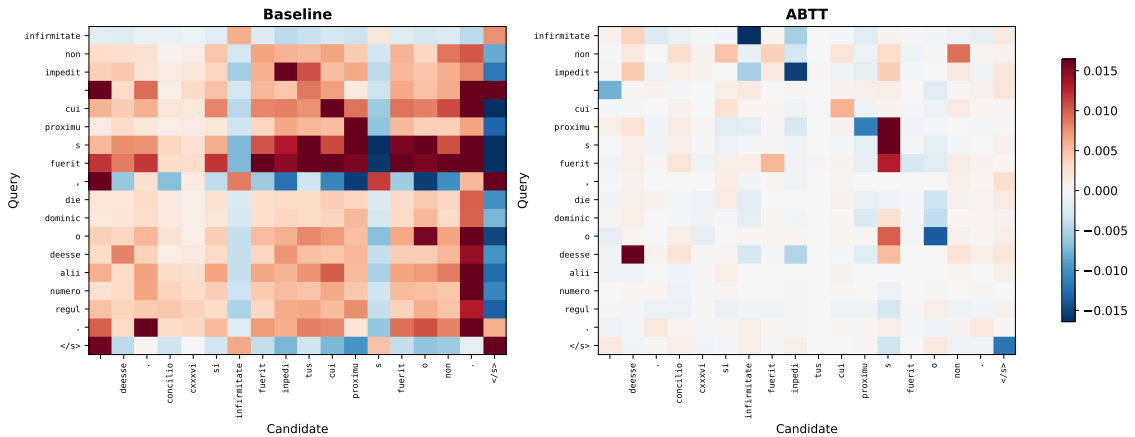


Figure 5: Derived pairwise heatmap for one PhilTa retrieval case. The left panel shows the baseline representation and the right panel shows the same example after ABTT. Both axes use original decoded tokens. Cell values combine token-token cosine similarity with single-sequence integrated-gradient scores.

Model	Method	$\rho_{\text{LOO}} \uparrow$		Suff@10% $\uparrow$		Suff@25% $\uparrow$		Suff@50% $\uparrow$		Comp@10% $\uparrow$		Comp@25% $\uparrow$		Comp@50% $\uparrow$		MinFrac@0.70 $\downarrow$		MinFrac@0.80 $\downarrow$		MinFrac@0.90 $\downarrow$		MinFrac@0.95 $\downarrow$	
		base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt
LaTa	BERTScore	0.282	0.496	-0.019	0.565	-0.032	0.788	0.016	0.960	0.292	0.074	0.298	0.192	0.282	0.426	0.377	0.210	0.403	0.297	0.436	0.433	0.467	0.517
	OT	0.031	0.382	0.929	0.581	0.961	0.820	0.979	0.974	0.872	0.071	0.882	0.179	0.827	0.361	0.087	0.184	0.115	0.258	0.160	0.376	0.197	0.464
	Att-weighted	-0.014	0.001	-0.185	0.265	-0.216	0.459	0.341	0.709	0.015	0.046	0.022	0.135	0.182	0.255	0.334	0.476	0.347	0.584	0.377	0.705	0.408	0.787
	Att-standalone	0.144	0.575	0.788	0.721	0.910	0.905	0.962	1.010	1.737	0.108	0.954	0.275	1.074	0.553	0.116	0.132	0.172	0.209	0.273	0.341	0.322	0.436
	DLA	0.075	0.601	0.973	0.704	1.009	0.893	1.035	0.989	1.200	0.123	1.250	0.292	1.251	0.555	0.086	0.128	0.118	0.208	0.154	0.353	0.192	0.452
	random	0.004	-0.002	0.265	0.547	0.519	0.736	0.819	0.871	0.026	0.021	0.030	0.053	0.177	0.136	0.183	0.220	0.219	0.330	0.288	0.514	0.338	0.659
	inverse	-0.003	-0.370	-0.202	0.276	-0.179	0.436	-0.205	0.653	-0.007	-0.017	-0.013	-0.022	-0.015	0.035	0.615	0.513	0.706	0.645	0.811	0.797	0.847	0.883
PhiTa	BERTScore	0.561	0.646	0.695	0.841	0.874	1.016	0.984	1.105	0.047	0.231	0.113	0.442	0.253	0.686	0.130	0.122	0.199	0.178	0.311	0.254	0.393	0.310
	OT	0.340	0.616	0.641	0.846	0.818	0.971	0.940	1.040	0.043	0.289	0.114	0.523	0.233	0.749	0.152	0.084	0.237	0.136	0.401	0.221	0.541	0.289
	Att-weighted	0.111	0.041	0.589	0.355	0.807	0.592	0.923	0.775	0.013	0.050	0.051	0.156	0.145	0.311	0.187	0.376	0.268	0.485	0.415	0.630	0.550	0.723
	Att-standalone	0.287	0.217	0.666	0.486	0.822	0.705	0.937	0.888	0.022	0.095	0.073	0.240	0.228	0.458	0.148	0.270	0.240	0.358	0.405	0.479	0.533	0.560
	DLA	0.531	0.599	0.729	0.872	0.862	0.970	0.959	1.006	0.068	0.311	0.159	0.567	0.337	0.776	0.102	0.069	0.182	0.115	0.345	0.205	0.486	0.285
	random	-0.004	-0.002	0.650	0.400	0.815	0.603	0.922	0.792	0.011	0.034	0.031	0.093	0.084	0.220	0.150	0.351	0.242	0.478	0.412	0.642	0.576	0.751
	inverse	-0.329	-0.614	0.514	0.046	0.636	0.109	0.774	0.255	0.002	-0.030	0.015	-0.049	0.065	-0.038	0.325	0.859	0.516	0.909	0.730	0.951	0.837	0.970
mT5-base	BERTScore	0.351	0.569	0.906	0.777	0.968	1.059	0.991	1.159	0.004	0.227	0.012	0.486	0.036	0.790	0.030	0.178	0.051	0.244	0.104	0.315	0.187	0.356
	OT	0.151	0.678	0.840	0.908	0.933	1.127	0.976	1.205	0.004	0.313	0.012	0.641	0.039	0.996	0.045	0.127	0.086	0.178	0.183	0.252	0.321	0.305
	Att-weighted	0.242	0.264	0.937	0.962	0.974	1.067	0.990	1.075	0.004	0.319	0.011	0.589	0.035	0.845	0.015	0.085	0.028	0.137	0.068	0.228	0.148	0.292
	Att-standalone	0.145	0.268	0.886	0.616	0.960	0.760	0.986	0.857	0.004	0.221	0.011	0.412	0.035	0.587	0.032	0.270	0.060	0.374	0.124	0.499	0.220	0.586
	DLA	0.237	0.552	0.901	0.934	0.950	1.051	0.979	1.032	0.004	0.341	0.014	0.622	0.046	0.825	0.024	0.083	0.045	0.133	0.110	0.231	0.245	0.312
	random	0.002	0.002	0.917	0.350	0.968	0.534	0.989	0.741	0.001	0.048	0.004	0.119	0.011	0.268	0.028	0.401	0.046	0.537	0.095	0.697	0.175	0.785
	inverse	-0.143	-0.686	0.687	-0.201	0.884	-0.186	0.962	0.002	0.002	-0.092	0.007	-0.168	0.025	-0.210	0.114	0.869	0.158	0.912	0.260	0.953	0.408	0.974

Table 21: Supplemental attribution sweep for additional methods and sanity baselines. *random* and *inverse* are diagnostic baselines, not paper-facing attribution methods. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at $\{10\%, 25\%, 50\%\}$; MinFrac is the minimum retained query-token fraction needed to recover $\tau \in \{0.70, 0.80, 0.90, 0.95\}$ of the full cosine (lower is better). Ratio metrics require $|S_v(q_{\text{full}}, c)| \geq 0.05$; ρ_{LOO} excludes $|\Delta| < 10^{-6}$. The completeness report contains no missing metric cells.

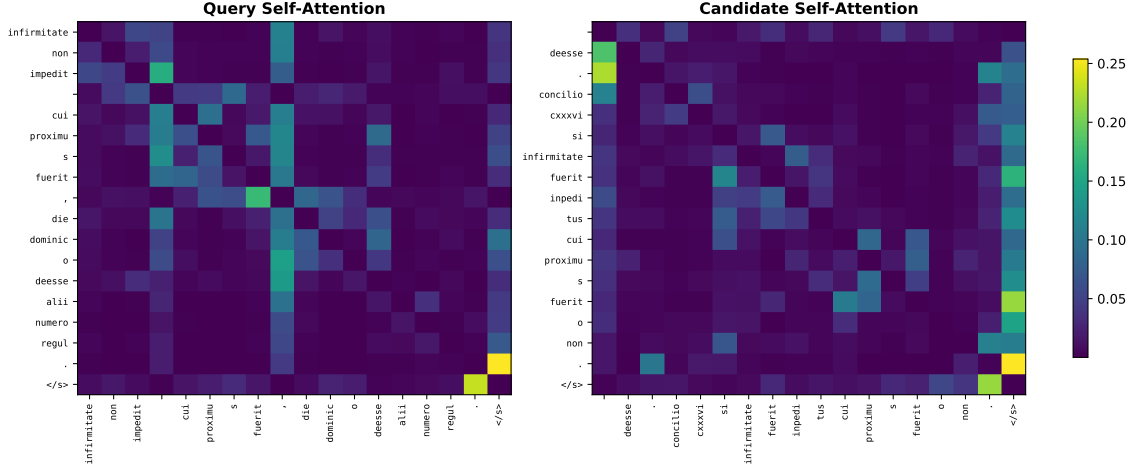


Figure 6: Companion self-attention maps for the same PhiTa example as Figure 5. ABTT is applied after encoding, so attention is included only as supporting context.

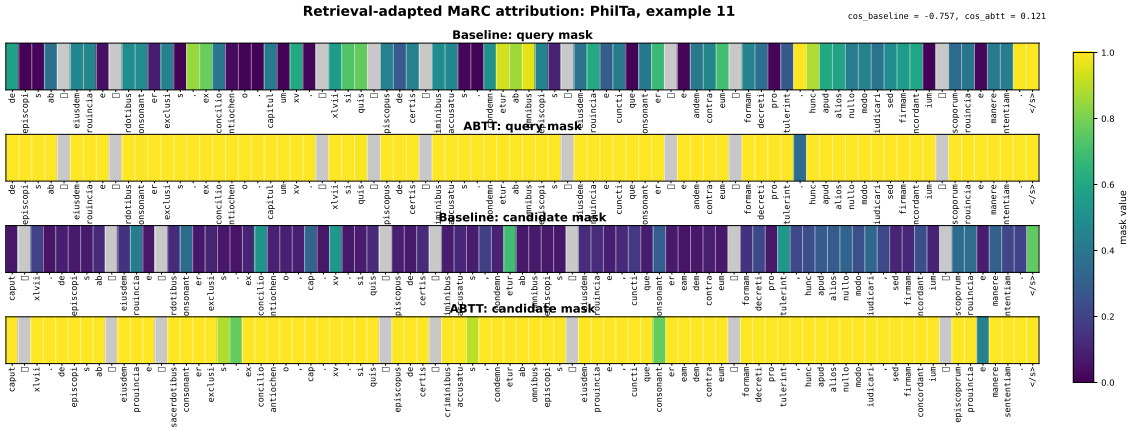


Figure 7: Retrieval-adapted MaRC masks for the same PhiTa pair as Figure 5. Top row: mask on the query with the candidate fixed. Bottom row: mask on the candidate with the query fixed. Left column: baseline cosine target. Right column: ABTT-corrected cosine target.

2D TSNE of labelled fragments at each model's train-selected layer

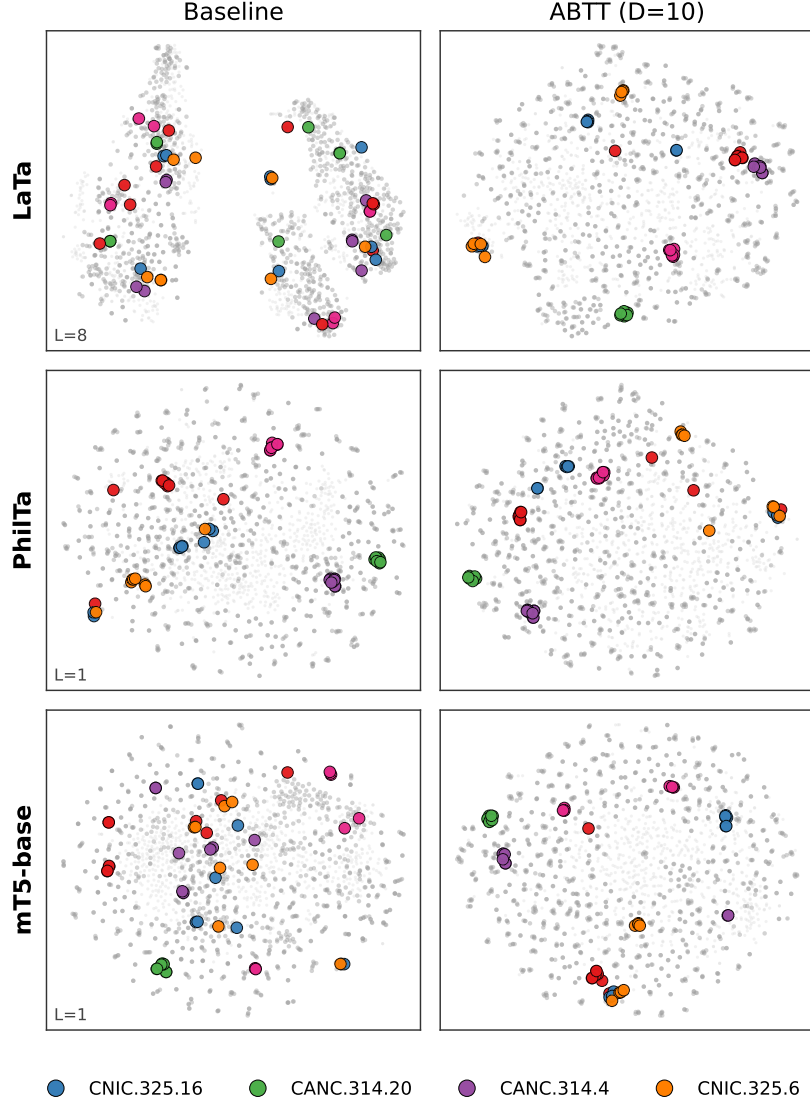


Figure 8: t-SNE projections of all 1,705 manuscript witnesses for LaTa, PhilTa, and mT5-base, each at the layer chosen on the train split: the layer with the highest training-set directory accuracy at rank 1 under ABTT, printed in each panel as L and equal to the Task B ABTT layer in Table 6. Columns compare baseline mean pooling with ABTT-only using $D = 10$, fit on the training embeddings. Colors mark the six largest directories; other witnesses of directories with at least two witnesses are mid gray, and singletons light gray. The silhouette scores in Section 6 are computed in these two-dimensional coordinates over the 1,160 winnable witnesses, those whose directory has at least two witnesses.

2D UMAP of labelled fragments at each model's train-selected layer

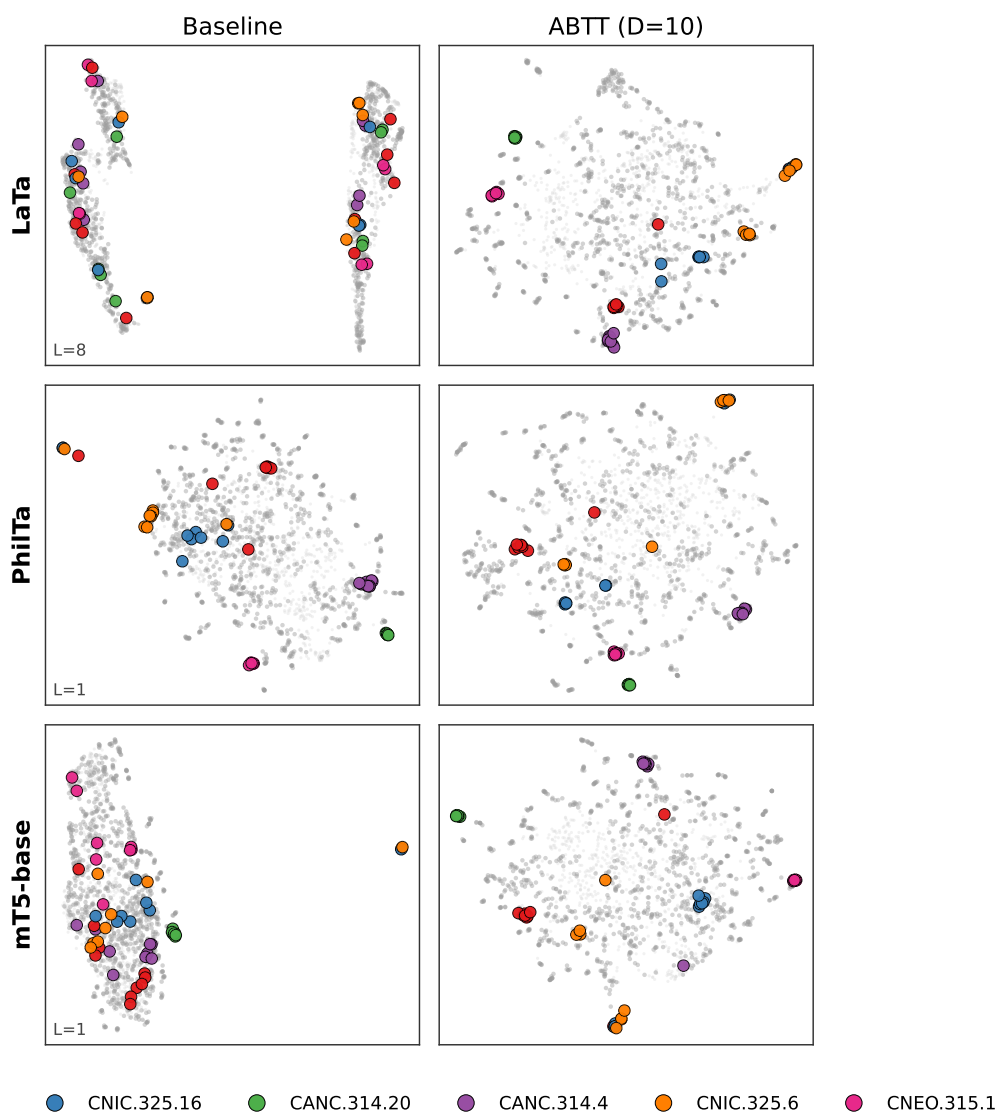


Figure 9: UMAP projections with the same configuration as Figure 8. The same cluster-tightening pattern appears under a different projection method.

2D TSNE of labelled fragments at each model's train-selected layer

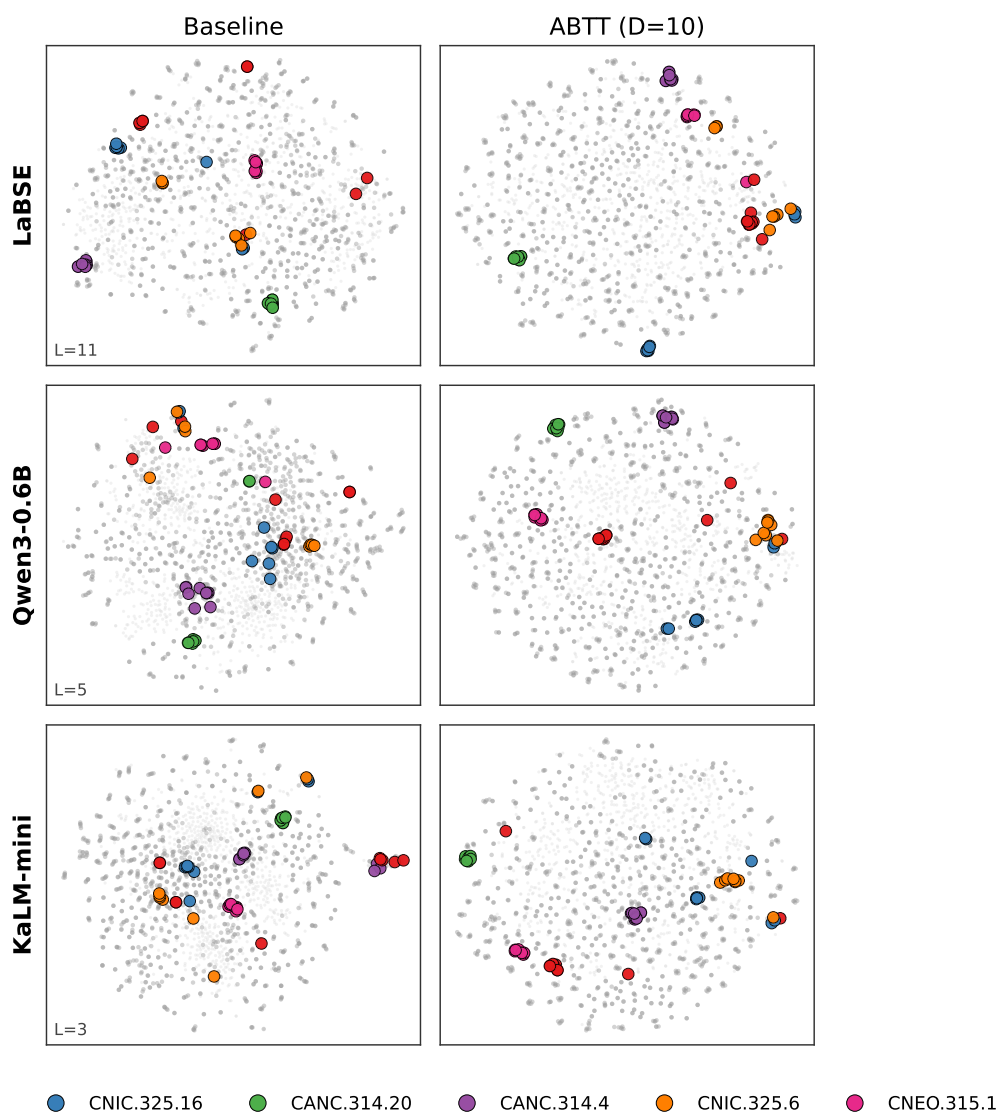


Figure 10: t-SNE projections of all 1,705 manuscript witnesses for LaBSE, Qwen3-0.6B, and KaLM-mini under the layer rule of Figure 8: the layer with the highest training-set directory accuracy at rank 1 under ABTT, printed in each panel and equal to the Task B ABTT layer in Table 6. Columns compare baseline mean pooling with ABTT-only using $D = 10$. Colors mark the same six largest directories as Figure 8.

2D UMAP of labelled fragments at each model's train-selected layer

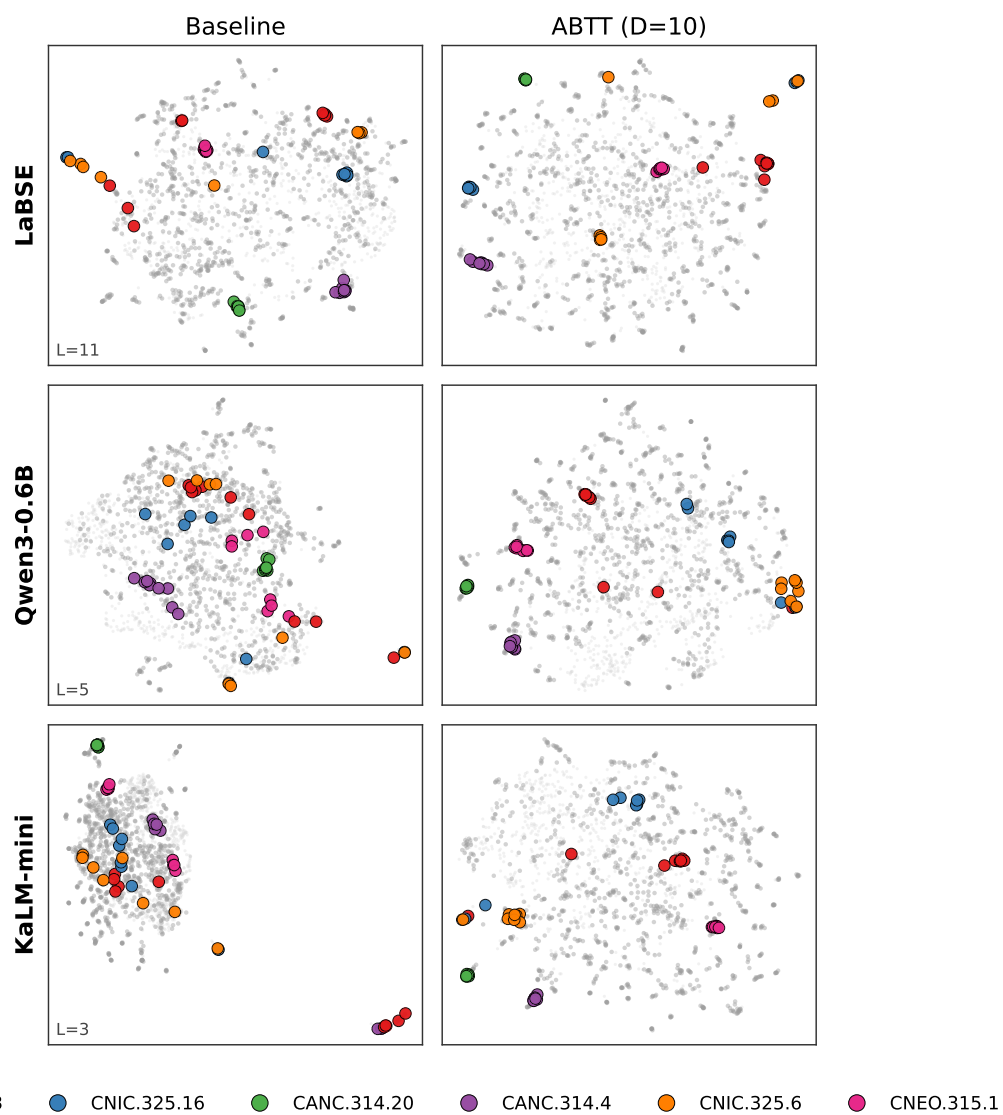


Figure 11: UMAP projections matching Figure 10. The cluster-tightening pattern appears under a second projection method.